KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY

COLLEGE OF ENGINEERING

DEPARTMENT OF AGRICULTURAL ENGINEERING

AE 196

ENGINEERING WORKSHOP PRACTICE

LECTURE NOTES

Lecturer:
Dr. Kojo Atta Aikins
Department of Agricultural Engineering

Course Outline

1. An Introduction to Technical Report Writing
2. Study of Hand Tools, Fitting Tools, & Power tools.
3. Machine Tool Work using Working Drawings & Metrology:
    a. Turning,
    b. Milling and
    c. Shaping.
4. Machine Building Exercises.
5. Welding: Gas & Electric Exercises.
6. Sheet Metal Work exercises.

Recommended Text Books

1. Element Of Workshop Technology by S. K. Hajra Choudhury and A. K. Hajra Choudhury
2. All About Machine Tools by Heinrich Gerling
3. Workshop Technology by H. S. Bawa
4. Machining Fundamentals Instructor's Resource
5. M.P. Groover, "Fundamentals of modern manufacturing, 3rd edition, (2007).
6. The Hong Kong Polytechnic University (2012). Marking Out, Measurement, Fitting & Assembly.

1.0 HAND TOOLS

"Man without Tools is nothing; with tools he is all." - This sentence, which was defined by Thomas Carlyle, has well elaborated the importance of tooling to a man.

The term 'Tooling' as applied to the engineering discipline refers to any equipment or instruments that give helps in the production of a product or any related activities. Simply speaking, it ranges from the most fundamental type of hand tools, such as a File, to the very complex machine tools, such as a computer numerical control (CNC) Machining Centre.

Thus, one may ask the question - Why we still have to use hand tools in this modern age of technologies?

Yes, it is reasonable to say that the efficiency of any hand processes is low and the quality of outcome depends highly upon the skill of individuals. Perhaps it is fair to consider the following points before a definite answer is given to the above question: -

Accuracy: Although the CNC machine can give a higher degree of dimensional accuracy when compared with the inconsistent outcome of hand fitting, the extreme high degree of flatness required for a surface table or a machine slideway is usually obtained by hand scraping only.

Flexibility: Hand processes are very flexible and can be carried out at any place necessarily while machining processes cannot be. In addition, machining usually requires a rigid setting up, while fitting is simple.

Quantity: For large batch size, advanced production machines are commonly employed in order to maintain the accuracy as well as the efficiency, but for "jobbing type" works, such as manufacturing of a prototype or repairing a single component, it would be uneconomic to use these advanced machine tools. Instead, "jobbing type" works are usually produced by conventional machining and followed by hand fitting necessarily.

Final Assembly: In the assembly of precise component parts, no matter how accurate they are being produced, a skilful fitter is often required to give the necessary "finishing touch" on them to ensure that everything goes together correctly.

1.1 Measuring Tools in Workshop

1.1.1 Steel Rule

It is a stiff, straight steel strip having all the faces machined true. On one of the flat faces, graduations are marked in inches and centimetres. The least count is 1mm. It is used to set out dimensions.

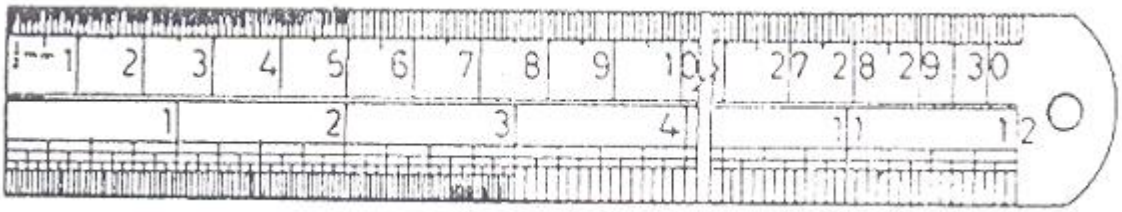


Figure 1-1: Steel rule

1.1.2 Callipers

Callipers are very simple tools used together with a steel rule for the measurement or comparison of linear dimensions. An experienced worker can achieve ±0.05mm in the measurement. Callipers are classified into two types:

1.1.2.1 Outside Callipers

Outside callipers, as shown in Figure 1-2, are used for measuring external dimensions such as the length, diameter, or even the thickness of a solid.

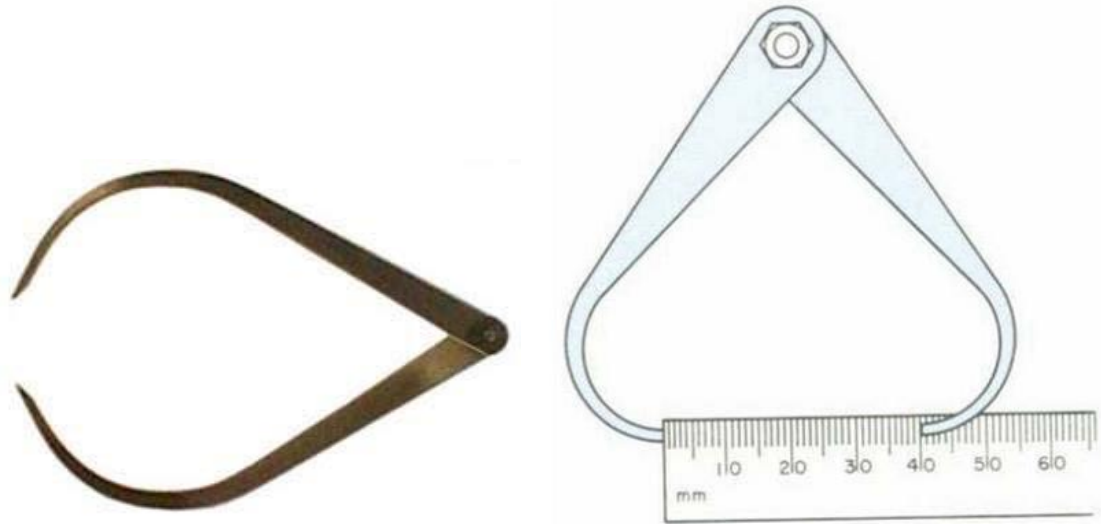


Figure 1-2: Outside Callipers

1.1.2.2 Inside Callipers

Inside callipers, as shown in Figure 1-3, are used for measuring internal dimensions such as the diameter of a hole, or the width of a slot, etc.



Figure 1-3: Inside Callipers

1.1.3 Vernier Callipers

Vernier Callipers, as shown in Figure 1-4, are more precise tools that are capable for measuring external dimensions, internal dimensions, and depths. Both pairs of measuring jaws and the depth gauge have the main features that including a main scale and a vernier scale.

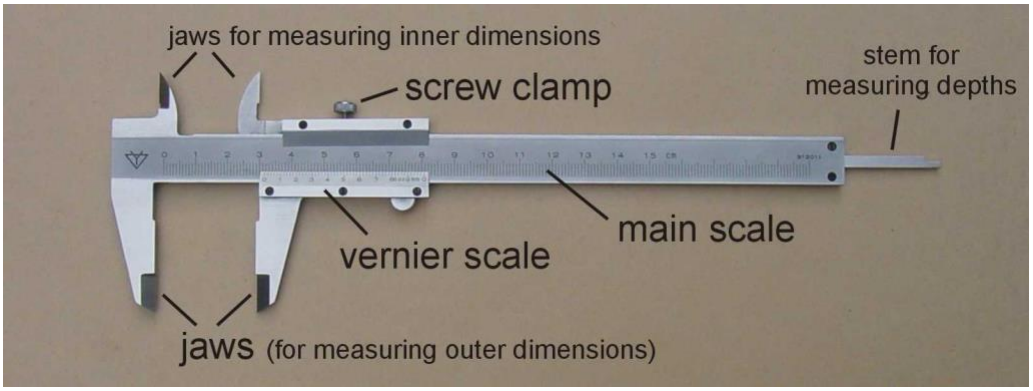


Figure 1-4: Vernier Callipers

The resolution of a vernier scale is determined by the difference on the distance of one division on the main scale and one division on the vernier as shown in Figure 1-5. For example, a vernier scale of length 49mm is divided into 50 equal divisions. That means ONE division on the vernier represents 49/50 = 0.98mm while ONE division on the main scale represents 1mm. Then, the resolution of the vernier is 1mm - 0.98mm = 0.02mm.



Figure 1-5: Vernier Reading



Figure 1-6: Vernier Height Gauge

1.1.4 Vernier Height Gauge

A vernier height gauge, as shown in Figure 1-6, is used for measuring height of an object or marking lines onto an object of given distance from a datum base.

1.1.5 Micrometre

A micrometre is a more precise measuring instrument than vernier callipers. The accuracy comes from the fine thread on the screw spindle. The ratchet prevents excess force from being applied. Generally, the screw spindle has a pitch of 0.5mm. The thimble is divided into 50 equal divisions. Common types of micrometres used in the workshops are:

1.1.5.1 Outside Micrometre

An outside micrometre, as shown in Figure 1-7, is used for measuring external dimensions. The work to be measured is placed between the anvil and the tip of the spindle.

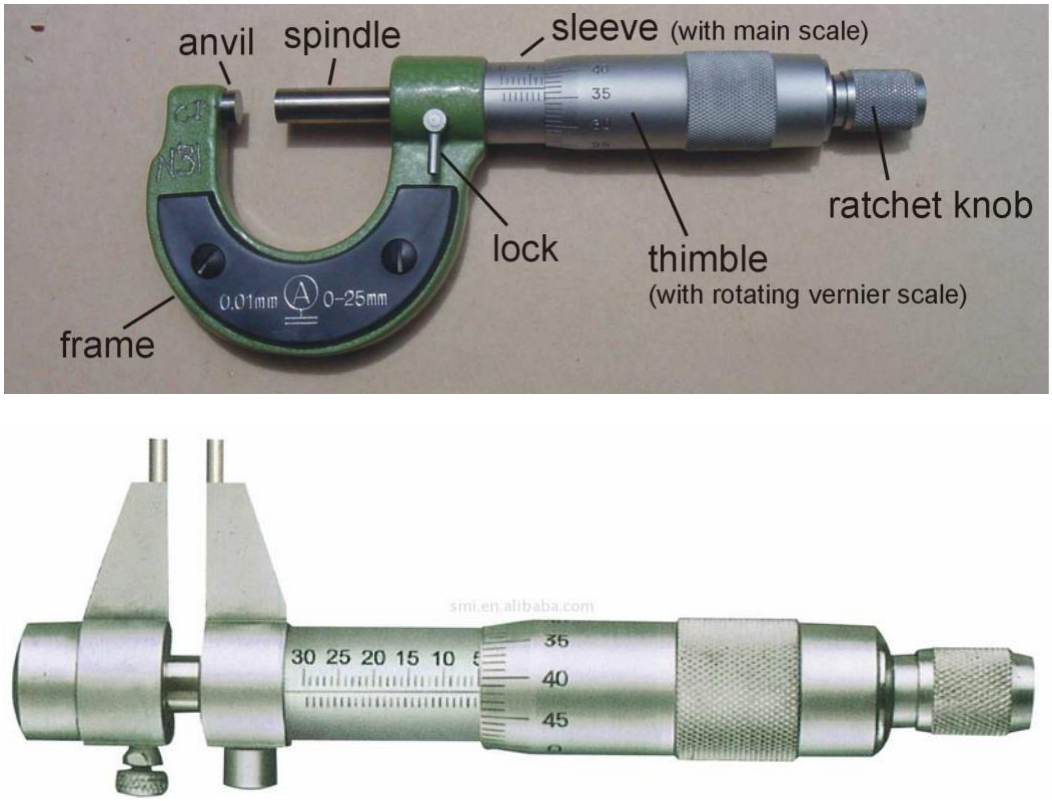


Figure 1-7: Outside Micrometre (top) and Inside Micrometre (bottom)

1.1.5.2 Inside Micrometre

The structure of this micrometre is as similar as that of an outside micrometre. It is used for measuring internal dimensions as shown in Figure 1-7.

1.1.5.3 Depth Micrometre

A depth micrometre, as shown in Figure 1-8, is used for measuring the depth of a hole, slot and keyway, etc. A complete set of depth micrometre is equipped with spindles of different lengths, which can be interchanged to suit different measuring ranges.



Figure 1-8: Depth Micrometre

1.1.6 Protractor

1.1.6.1 Engineer's Protractor

An engineer's protractor, as shown in Figure 1-9, is a general purpose tool used for the measuring/ checking of angles, e.g. the angle of drill head, angle of cutting tool, and even for the marking out of angles on a component part.

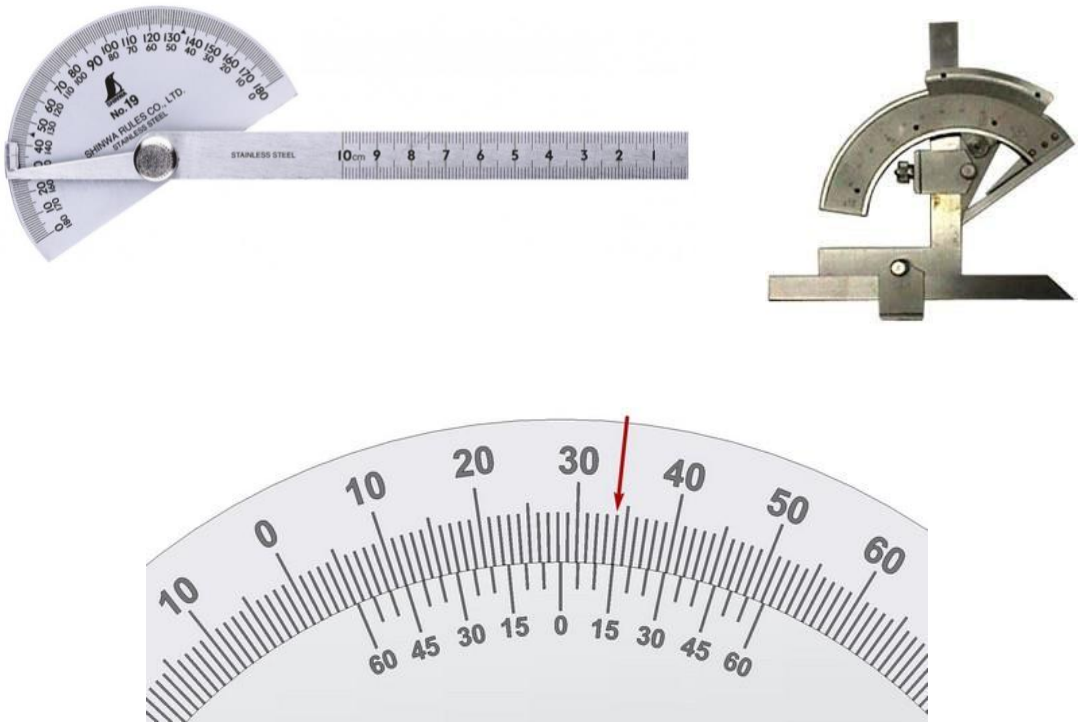


Figure 1-9: Engineer's Protractor (top left), Vernier Protractor (top right) and a reading on a vernier protractor (bottom)

1.1.6.2 Vernier Protractor

This is a precision measuring tool whose accuracy of measurement can reach ±5 minutes of an angle through the vernier scale as shown in Figure 1-9.

1.1.7 Combination Set

A combination set, as shown in Figure 1-10, is a set of equipment combining the functions of protractor, engineer square, steel rule, centre finder, level rule, and scriber.



Figure 1-10: Combination Set

1.1.8 Dial Indicator

A dial indicator (dial gauge) can measure dimensions up to an accuracy of 0.01mm or even less. The principle of it is that the linear mechanical movement of the stylus is magnified and transferred to the rotation of pointer as shown in Figure 1-11. It is usually used as a comparator for calibration or alignment of machine.



Figure 1-11: Dial Indicator

1.2 Marking Out Tools in Workshop

Marking out is the preliminary work of providing guidance lines and centres before cutting and machining. The lines are in 3-D and full-scale. The workpiece can then be cut or machined to the required shape and size. The common tools used for marking out are as follow:

1.2.1 Scriber

A scriber, as shown in Figure 1-12, is used for scratching lines onto the workpiece. It is made of hardened tool steel, which is hardened and tempered to a suitable hardness.

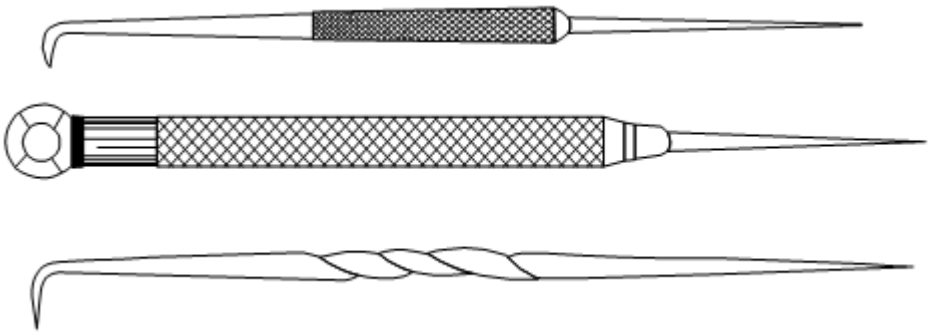


Figure 1-12: Scribers

1.2.2 Engineer's Square

Engineer's square, as shown in Figure 1-13, is made of hardened tool steel. It is ground to a precise squareness and straightness. It is not only used for checking the straightness and the squareness of a workpiece, but it can also be used for marking parallel or perpendicular lines onto a workpiece.

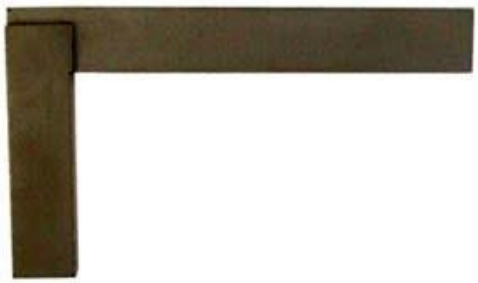


Figure 1-13: Engineer's Square

1.2.3 Spring Dividers

Spring dividers, as shown in Figure 1-14, are made of hardened tool steel. The legs are used for scribing arcs or circles onto a workpiece.



Figure 1-14: Spring Dividers

1.2.4 Punch

There are two types of punch named as the Centre Punch and the Dot Punch. A dot punch has a point angle of either 30 or 60°, so it can be used for marking small dots on the reference line. The centre punch has a point angle of 90° as shown in Figure 1-15. It is used for making a large indent on a workpiece for drilling. Both punches are made of hardened tool steel.



Figure 1-15: Punch (Dot and centre)

1.2.5 Surface Plate

Surface plate, as shown in Figure 1-16, is made of malleable cast iron. It has been machined and scraped to a high degree of flatness. The flat surface is used as a datum surface for marking out and measuring purposes. Large surface plate that can stand on the floor is known as surface table.



Figure 1-16: Surface Plate

1.2.6 Angle Plate

An angle plate, as shown in Figure 1-17, is used for supporting or setting up work vertically. Slots and holes are provided for mounting and fixing the workpiece. It is made of cast iron and ground to a high degree of accuracy.

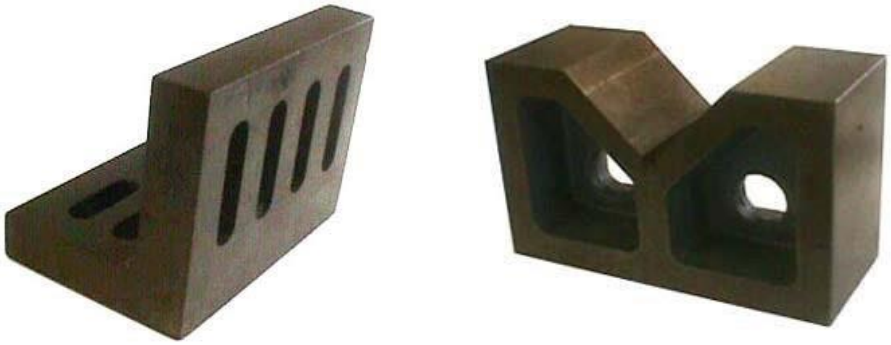


Figure 1-17: Angle Plate (left) and Vee Block (right)

1.2.7 Vee Block

Vee blocks, as shown in Figure 1-17, usually in a couple are made of cast iron or steel in case-hardening. They are generally used to hold circular workpiece for marking out or machining.

1.3 Hand Tools for Workshop

1.3.1 Bench Vice

A bench vice, as shown in Figure 1-18, is a device for holding the workpiece for carrying out most hand processes. The body of the vice is made of cast iron while the two clamping jaws are made of hardened tool steel. Some bench vice has a swivel base, which can set the workpiece at an angle to the table. The vice height should be correct ergonomically. In order to protect the workpiece from being damaged by the jaws, vice clamps are employed. Vice clamps, which are made of copper, are fitted over the vice jaws when holding finished work.



Figure 1-18: Bench Vice

1.3.1.1 Care of Vices

a. Do not make direct impact on the vice body with a hammer.

b. Light hammering can be done on and only on the anvil of the vice.

c. To avoid over clamping, the handle of the vice should be tightened by hand only

1.3.2 Files

Files are the most important hand tools used for the removal of materials. They are made of hardened high carbon steel with a soft 'tang', which is used for fixing a handle. Files are categorised according to the following:



Figure 1-19: File

- Length - measured from the shoulder to the tip
- Shape - the cross-sectional profile
- Grade - the spacing and pitch of the teeth
- Cut - the patterns of cutting edge

1.3.2.1 Save Edge

There are no cutting edges on one side of the hand file. The purposes for the save edge is to avoid the worker from damaging the work, when he is filing a shoulder position.

1.3.2.2 Shape of Files

Hand File - The common file used for roughing and finishing. It is rectangular in cross section and parallel in width. It has **double cut teeth** on both faces, single cut teeth on one edge, and one save edge.



Figure 1-20: Hand File

Flat File - It is similar to a hand file, which is rectangular in cross section and tapered slightly in width towards the tip. It has double cut teeth on both faces and single cut teeth on both edges.



Figure 1-21: Flat File

Half-round File – The cross section is a chord of circle and it tapers towards the tip. It is used for forming radii, grooves, etc. Its flat side is used for finishing flat surfaces.



Figure 1-22: Half-round File

Round File - This is a circular cross sectioned file which tapers toward the end. It is used for enlarging holes and producing internal round corners. It usually has double cut teeth in the larger sizes, and single cut teeth for the smaller sizes.



Figure 1-23: Round File

Square File - This is a file with square in section and taper towards the tip. It usually has double cut teeth on all four faces. It is used for filing rectangular slots or grooves.



Figure 1-24: Square File

Three Square File - It is also known as triangular file. It is triangular in cross section with taper towards the tip. It has double cut teeth on both faces. It is used for filing corners or angles less than 90°.



Figure 1-25: Triangular Square File

Needle Files - Needle files are a set of small files with their shapes made in a way similar to the large ones. They are generally used for small and delicate works, such as the repair of small instruments.

1.3.2.3 Cut Pattern

Single Cut - There is only one set of cutting teeth on the surface. It gives a less efficient cutting but a better finish. It is suitable for filing soft metal.

Double Cut - A double cut file has two sets of teeth, one at 70 degrees to the edge and the other one at 45 degrees to another edge. Thus, it is more efficient in cutting. It is easy to clog the teeth when it is used to file soft metals.

Rasp – These are very coarse teeth, like the nail. This type of teeth is commonly used for cutting off soft materials, such as rubber, PVC, and wood, etc.

1.3.2.4 Grade

It refers to the pitch (spacing) of the teeth that spread throughout the whole length of the file. Files with a rougher grade of cut give a faster metal removal rate but a poorer surface finish or the vice versa. It should be noted that, for the same grade of cut, a longer file would have a coarser pitch than a shorter one. There are five grades of cut, which are Rough, Bastard, Second, Smooth, and Dead Smooth.

The following three grades of cut are used commonly.

Bastard cut - medium teeth for general purposes, especially suitable for mild steel.

Second cut - finer teeth for cutting hard metals.

Smooth cut - fine teeth for finishing.
Filing is a hand process of using appropriate files to produce required shape, size and quality of surfaces or components.

1.3.2.5 Safety and Care of Files

1. File teeth are brittle and therefore file should be placed properly and should not be stacked on other tools.

2. New files should never be used on hard materials, e.g. casting or welding joint.

3. Some brittle metal, e.g. brass, is not readily filed with the worn teeth.

4. The pinning file should be cleaned regularly with file card / wire brush.

5. Files should be used in cutting on the forward stroke.

6. Files without handle should not be used.

1.3.3 File Card

When filing the soft metals, the small pieces of metal will tend to clog the teeth. If the file is not cleaned, the small pieces of metal will scratch on the surface of the work. It is called pinning. This case is frequently appeared when applying a new smooth file on the soft metals. The pinning can be removed with a file card as shown in Figure 1-26, which is a wire brush mounted on a block of wood. Sweep the file card along the grooves on the file until the pinning is removed.



Figure 1-26: Iron brush

1.3.4 Hacksaw

A hacksaw is generally used for cutting a metal into pieces. It consists of a frame and a saw blade as shown below. It is a "U" shaped steel frame with a pistol handgrip and a saw blade as shown in Figure 1-27. The frame, which is used for holding the blade firmly in position, is usually made of mild steel. The fixed type of frame can only take one length of blade, whereas the adjustable type can take different blade lengths. It has a wing nut to adjust the tension of the blade.



Figure 1-27: Hacksaw

1.3.4.1 Saw Blade

Saw blades are made of high carbon steel, alloy steel or high speed steel. They are supplied according to material, hardening, length and pitch.

Hardening – The saw blade is usually supplied with all-hard or flexible grade. The all-hard is very brittle, and it is suitable for the skilful user only. The flexible grade is tough, so it can twist an angle. It is suitable for cutting a curve or for the beginner to use.

Material – Usually, the saw blade is supplied with high carbon steel (HCS) and high speed steel (HSS). The HCS will be annealed from the heat generated by fraction of cutting. The HCS saw blade will lost its hardness when cutting the hard metal. The HSS can keep its hardness unless improper use.

Pitch - It is classified according to the number of teeth per 25mm.

Coarse blade (18T) is most suitable for soft material and thick workpiece.

Medium blade (24T) is suitable for steel pipe.

Fine blade (32T) is suitable for the thin metal sheet and thin copper pipe.



Figure 1-28: Pitches of Saw Blade

For safety, it is advised to keep at least 3 teeth of the blade, stand on the workpiece.

Length - The length of the blade is determined by the distance between the outside edges of the holes, which fit over the pegs.

Set - The teeth have a "set" to either side alternately, which causes the blade to cut a slit wider than the thickness of the blade, to prevent jamming.

1.3.4.2 Safety and Care of Hacksaw

1. The cutting action is carried on the forward action only. So the blade must be mounted with its teeth pointing forward.
2. Suitable tension should be applied on the blade to avoid breakage or loosen.
3. Change the blade if some teeth are broken.
4. Avoid rapid and erratic strokes of cut.
5. Avoid too much pressure.
6. Workpiece must be hold firmly.

1.3.5 Hammer

The type most commonly used is the ball peen hammer, which has a flat striking face and a ball-shaped end (call the peen). Hammer heads are made from medium carbon steel. The two ends must be hardened and tempered, the centre of the head with the eye is being left soft. It is specified according to its weight.



Figure 1-29: Ball peen hammer

1.3.5.1 Safety and Care of Hammer

1. The hammer head is firmly fixed to the shaft by a wedge.
2. The striking face of the hammer head does not wear.

1.3.6 Chisels

The most common type of chisel is the flat chisel. You should always wear safety goggles when working with chisels.

When using hammers and chisels, safety goggles should always be worn.



Figure 1-30: Flat chisel

This is the most common kind of chisel - a flat chisel. It's made of high-quality steel, and the end is tempered and hardened because it has to be harder than any of the metals you're likely to want to cut with it. The head of the chisel needs to be softer so it won't chip when it's hit with a hammer.



Figure 1-31: Crosscut chisel

This is a cross-cut chisel. It's called 'cross cut' because the sharpened edge is across the blade width. This chisel narrows down along the stock, so it's good for getting in grooves. It’s used for cleaning out or even making key-ways. The flying chips of metal should always be directed away from the user.



Figure 1-32: Gasket scraper

The gasket scraper is not a true chisel. It has a hardened, sharpened blade and it’s designed to remove a gasket without damaging the sealing face of the component. The scraper should be kept sharp to make it easy to remove all traces of the old gasket and sealing compound.

1.3.7 Screw-drivers

The correct screw-driver to use depends on the type of slot or recess in the head of the screw or bolt, and how accessible it is.

The correct screwdriver to use depends on the type of slot or recess in the head of the screw or bolt, and how accessible it is.

Most screwdrivers can't grip as securely as spanners, so it's very important to match the tip of the screwdriver exactly with the slot or recess in the head of a fastener. Otherwise the tool might slip, damaging the fastener, and worse still, you.

When using a screwdriver, always check where the screwdriver blade can end up if it slips off the head of the screw. A screwdriver can't tell the difference between a piece of steel, and a piece of you.

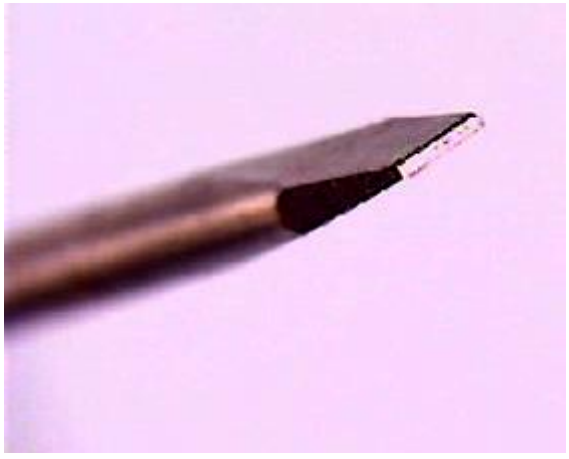


Figure 1-33: Blade (flat) screw driver

The most common screwdriver has a flat tip, or blade, which gives it the name, **blade screwdriver**. It's easy to see the blade should be almost as wide as the slot in the fastener so

that twisting force applied to the screwdriver is transferred right out to the edges of the head where it has most effect.

Not so easy to see, is that the blade should be a snug fit in the slot of the screw head. Then the twisting force is applied evenly along the sides of the slot. This guards against the screwdriver suddenly chewing a piece out of the slot, and slipping, just when most force is being exerted.

If viewed side-on, the blade should taper until the very end where the tip fits in the slot. If the tip of the blade isn't clean and square, it should be reshaped.

When you use a blade screwdriver, make sure you support the shaft with your free hand as you turn it. This helps keep the blade square on to the slot and centred. Slipping screwdrivers are a common source of damage and injury in workshops.

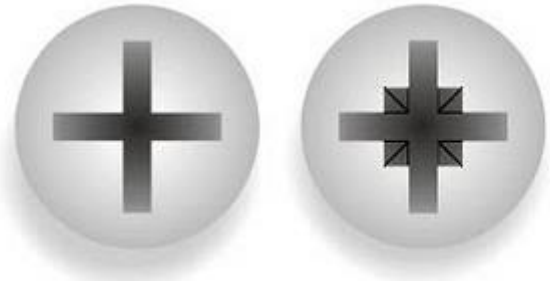


Figure 1-34: Phillips (left) and Pozidrive (right) screw heads

A screw or bolt with a **star-shaped recess** needs a **Phillips or a Pozidrive screwdriver**. The star-shaped slot holds the tip of the screwdriver securely on the head.



Figure 1-35: Phillips (left) and Pozidrive (right) screw drivers

The Phillips tip fits a tapered recess while the pozidrive fits into slots with parallel sides in the head of the screw. Both a Phillips and Pozidrive head can be pushed and twisted with more confidence, but again the screwdriver must be the right size. This is simplified with these 2 types of screwdriver because 4 sizes are enough to fit almost all fasteners with this sort of screw head.



Figure 1-36: Allen keys

The **Allen key** is designed to be a snug fit in screws with a socket head.

The socket and the key are hexagonal in shape, and there's a correct sized key for every socket, so Allen keys come in sets. They can come in either the metric or imperial system, and are categorized in millimetres or fractions of an inch, according to the distance across opposite flats of the hexagon. They give the best grip on a screw or bolt of all the drivers, and their shape makes them good at getting into tight spots.

1.3.8 Spanners

There are many types of spanners. The most common are the ring spanner, open end spanner and the combination spanner. A spanner will only do a job properly if it’s the right size for the nut or the bolt to be turned.

The size used to describe a spanner/wrench is the distance across the flats of the nut or bolt to be turned. There are two systems in common use, metric, in millimetres, and imperial, in inches. Each of the systems has a range of spanners/wrenches especially made for each one. The systems can be identified on the spanner/wrench by either a number for metric spanners, or a fraction followed by A/F.

There are many types of spanners/wrenches. Choosing the correct one to use to do the job usually depends on two things: How tight is the fastener? In other words how much force is going to be applied to it, and, how accessible is it? - How much room is there to get the spanner onto the fastener, and then turn it?

When being used, it is always possible a spanner/wrench will slip. Always try to anticipate what will happen if it does slip before putting a lot of tension onto it. If possible, always pull a spanner/wrench towards you rather than pushing it away.



Figure 1-37: Ring spanners

Ring spanners grip a fastener at the corners just like a socket spanner, just the sort of grip that is needed if a nut or bolt is very tight. Ring spanners have different sized heads at each end. They are not as convenient as sockets but can fit into places that a socket cannot. One disadvantage of the ring spanner is that it can be awkward to use once the nut or bolt has been loosened.



Figure 1-38: Open end spanner

Open end spanners/wrenches slip easily and quickly onto fasteners, and that's particularly important for nuts and bolts in awkward places. The angle on the head allows it to be used in two different positions. While an open-end spanner often gives the best access to a fastener, if it's extremely tight the open-end shouldn't be used as this type of spanner/wrench only grips across two flats. If the jaws flex slightly or the flats don't fit tightly between them, the spanner can suddenly slip when force is applied.

The best way is to use a ring spanner to break such a bolt or nut free, then the open-end. The open-end spanner should only be used on fasteners that are no more than firmly tightened.



Figure 1-39: Combination spanners

The **combination spanner/wrench** provides the user with the best features of each of the other types. It has a ring on one end for gripping and breaking the fastener's hold, and an open ended spanner of the same size on the other end.



Figure 1-40: Flare nut spanner

A variation on the open end head is the **flare nut spanner**. It gives a better grip because the flats meet on 5 sides, not 2. The open 6th side lets the spanner be used on nuts and fittings associated with pipes and tubing. Do not use the flare-nut spanner on extremely tight fasteners as the jaws may spread, damaging the nut.



Figure 1-41: Wrench (Shifting) / adjustable spanners

There is another type of universal adjustable open end spanner/wrench in use. Usually referred to as a **shifting spanner/wrench** or simply "shifters". The lower jaw can be moved to fit any fastener size within the spanner range. Shifting spanners should only be used if the correct sized spanner is not available. Both the fastener and spanner could be damaged if they are used on really tight bolts or nuts.

2.0 WELDING

Welding may be defined as a process of joining together metal pieces by interpenetration of its atoms. The process is carried out by heating the metal pieces to plastic or fusion state, bringing them together with or without application of pressure and filler material.

The metals being joined are locally melted and when solidified produce a solid mass. The joint strength is therefore as strong as the metal being joined. A filler rod, with a melting temperature similar to that of the metals being joined, is sometimes used to make up losses during welding, to fill any gap between the joint surfaces, and to produce a fillet.

In contrast, Soldering and Brazing do not involve melting the workpiece but rather a lower-melting-point material is melted between the workpieces to bond them together.

2.1 Welding Classifications

Welding processes are classified in two kinds: Plastic welding and Fusion welding. They are also subdivided into different types.

2.1.1 Plastic welding:

Forge, cold, pressure gas, resistance welding and thermit welding with pressure. Resistance welding has three types: Butt welding, seam welding and spot welding.

2.1.2 Fusion welding:

Gas and Arc welding, and thermit welding without pressure.

In order to melt metals to be joined by fusion, a source of high heat energy is required. The two chief methods of obtaining the high temperature required are by an electric arc and by a mixture of gases.

2.2 Gas welding

Gas welding is one of the fusion welding processes where the metal and the filler metal (welding rod) are heated to the melting point temperature. The heat for this method is obtained by burning a mixture of combustible gas and oxygen to provide an intense flame. The two gases are mixed in the correct proportion in the mixing chamber of the welding torch. Control of the gas mixture is by two valves in the handle of the torch. Interchangeable nozzles of different sizes are fitted to the touch to give a range of flame intensities to suit a variety of applications. The most widely used combustible gas is acetylene, which when mixed with oxygen gives a flame temperature of around 3250°C.

The flame is directed on to the metal being joined, resulting in a molten pool of metal. Filler metal is added when required and is usually the same material as that being joined. Fluxes may be required, as in the case of welding aluminium or copper alloys.

This method is used to weld a wide variety of metals. It is cheap and portable, making it invaluable for repair work. It can be used on steel up to 25 mm thick, but it is generally employed on sheet material 16 SWG (1.6 mm) and below.

2.2.1 Oxygen and Acetylene gas

Oxygen in pure form is mixed with acetylene (C_2H_2) produced by chemical reaction between water and calcium carbide. After mixing with pure oxygen, acetylene burns to produce a flame of temperature varying between 3200°C to around 3500°C depending upon the ratio of oxygen

and acetylene. The flame known as oxy-acetylene flame is obtained through the tip of a welding torch in two stages.

The chemical reaction in the first stage is given by the equation:

$C_2H_2 + O_2 = 2CO + H_2$

This is a bright white cone type of flame available close to the tip opening. This flame gives most concentrated heat with highest welding temperature.

In the second stage, the carbon monoxide and hydrogen produced in the first stage reacts further with oxygen of the surrounding air to the chemical equation:

$4CO + 2H_2 + 3O_2 = 4CO_2 + 2H_2O$. This flame is of a larger blue colour, surrounding the bright white flame obtained in the first stage. This is used to pre-heat the metal and to protect the mother metal from oxidation.

2.2.2 Oxy-acetylene flame

The flame produced during gas welding is classified in three groups depending upon the ratio of oxygen and acetylene.

- A neutral flame
- An oxidizing flame
- A carburizing flame

2.2.2.1 A Neutral flame

It is obtained when equal volumes of oxygen and acetylene are supplied. It has two cones, an inner white cone flame which develops heat for welding and an outer blue flame (enveloping the first) used to pre-heat the metal and protect oxidation. It is used to weld steel, stainless steel cast iron, copper, aluminium, etc.

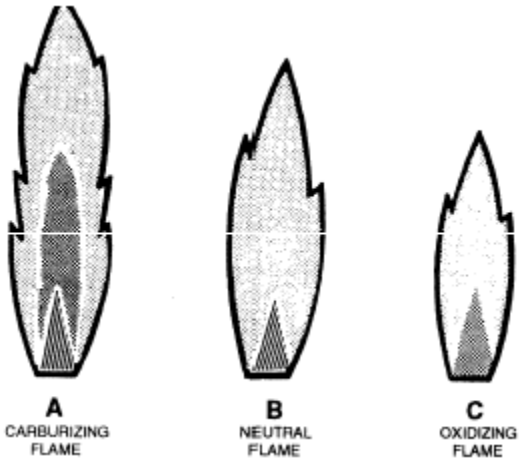


Figure 2-1: A) Carburizing, B) neutral and C) oxidizing flames

2.2.2.2 An Oxidizing flame

It is produced when more oxygen than required for chemical reaction is supplied. In appearance it resembles a neutral flame except that it has a shorter inner white cone flame. It is mostly used to weld brass.

2.2.2.3 A Carburizing flame

Carburizing flame is produced when more acetylene, than required for chemical reaction is supplied. It has three zones.

- an inner white cone flame
- an intermediate white feather, whitish in colour
- an outer blue flame

It has only a few applications, for example, hard facing, etc.

2.2.3 Oxy-acetylene welding torch

A welding torch mixes the gases in desired ratio and delivered the mixture at its tip to produce the flame. It is classified in two types:

- injector type
- positive pressure type

In the injector type, the high pressure oxygen passing through the venture creates a low pressure which sucks in the acetylene.

The positive pressure type works on equal pressure principles, where the oxygen and acetylene are supplied at equal pressure. The injector types are mostly used in industries. Torches are also classified according to its purpose, such as welding torch, cutting torch, brazing torch, etc. A welding torch can be worked both as an injector type and as a positive pressure type depending upon the design of the mixer.

2.2.4 Gas welding technique

The following procedures are adopted to produce a welded joint. The proper size of welding torch and filler material is selected. The welding rod has the same chemical composition as that of the parent metal. The diameter of rod is determined by the empirical formula:

$$d = \frac{s}{2} + 1$$

Where;

d = diameter of the rod (mm)
s = the thickness of the parent metal (mm)
The size of torch depends upon the thickness and conductivity of the work-piece. Metals having higher conductivity require a torch head of larger diameter than a metal of lower conductivity.

The properly prepared metal pieces are positioned. The acetylene valve is first opened and lighted with a friction spark lighter. The oxygen valve is then opened and the flame adjusted. The joint is the pre-heated. The welding rod is the held in the flame so that its end melts at about the same time as the base metal.

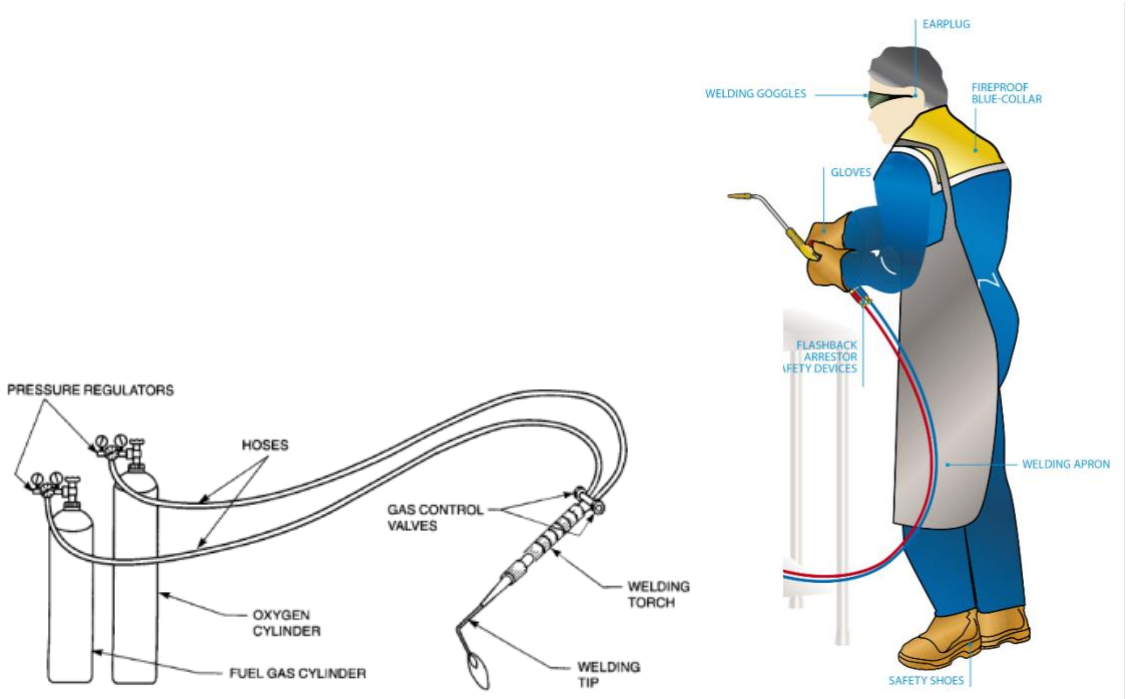


Figure 2-2: Equipment needed for gas welding

2.2.5 Equipment needed for Gas welding

- Welding torch
- Pressure regulators
- Hose and fittings
- Gas cylinders
- Goggles
- Gloves
- Spark lighter

2.3 Arc Welding

An arc welding is produced by passing an electric current between two electrode separated by a small gap. In arc welding, one electrode is the welding rod or wire, the other is the metal plate being joined.

The electrodes are connected to the electrical supply, one to the positive terminal and one to the negative. The arc is started by touching them and withdrawing the welding rod about 3 to 4 mm from the plate. When the two electrodes touch, current flows and as they are withdrawn, the current continues to flow in the form of spark. The resulting high temperature is sufficient to melt the metal being joined.

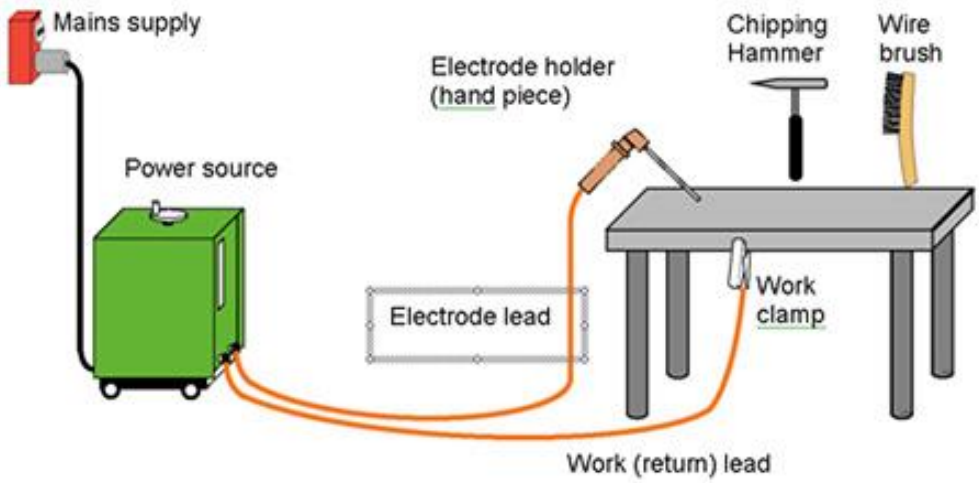


Figure 2-3: A set-up for electric arc welding

2.3.1 Welding Terminology

2.3.1.1 Filler Material

When welding two pieces of metal together, we often have to leave a space between the joint. The material that is added to fill this space during the welding process is known as the filler material (or filler metal). Two types of filler metals are commonly used in welding are welding rods and welding electrodes.

- **Welding Rod**: The term welding rod refers to a form of filler metal that does not conduct an electric current during the welding process. The only purpose of a welding rod is to supply filler metal to the joint. This type of filler metal is often used for gas welding.

- **Welding Electrode**: In electric-arc welding, the term electrode refers to the component that conducts the current from the electrode holder to the metal being welded. Electrodes are classified into two groups: consumable and non-consumable.

    - ***Consumable electrodes*** not only provide a path for the current but also supply filler metal to the joint. An example is the electrode used in shielded metal-arc welding.

    - ***Non-consumable electrodes*** are only used as a conductor for the electrical current, such as in gas tungsten arc welding. The filler metal for gas tungsten arc welding is a hand fed consumable welding rod.

2.3.1.2 Flux

Before performing any welding process, the base metal must be cleaned form impurities such as oxides (rust). Unless these oxides are removed by using a proper flux, a faulty weld may result. The term flux refers to a material used to dissolve oxides and release trapped gases and slag (impurities) from the base metal such that the filler metal and the base metal can be fused together. Fluxes come in the form of a paste, powder, or liquid. Different types of fluxes are available and the selection of appropriate flux is usually based on the type of welding and the type of the base metal.

2.4 Types of Welds

There are many types of welds. The most common types are the bead, surfacing, plug, slot, fillet, and groove.

2.4.1 Weld Bead

This is a weld deposit produced by a single pass with one of the welding processes – Figure 2-4 (a). A weld bead may be either narrow or wide, depending on the amount of transverse oscillation (side-to-side movement) used by the welder. A weld bead made without much weaving motion is often referred to as a stringer bead. On the other hand, a weld bead made with side-to-side oscillation is called a weave bead.



Figure 2-4: Weld Bead and Surfacing Welds

2.4.2 Surfacing weld

Several weld beads applied side-by-side are usually used in surfacing which is a welding process used to apply a hard, wear-resistant layer of metal to surfaces or edges of worn-out parts – Figure 2-4 (b).

2.4.3 Fillet Weld

A Fillet weld is triangular in shape and this weld is used to join two surfaces that are at approximately right angles to each other in a lap, tee, or comer joint (Figure 2-5).

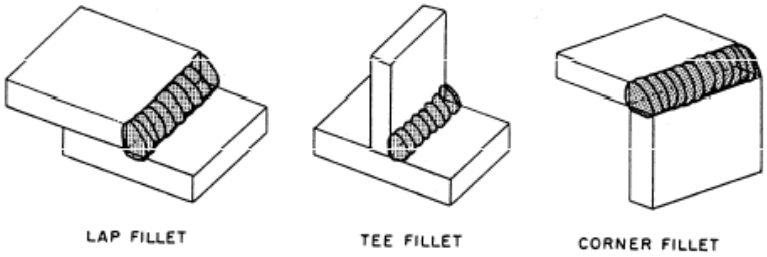


Figure 2-5: Types of Fillet Welds

2.4.4 Plug and Slot Welds

Plug and Slot welds are welds made through holes or slots in one member of a lap joint –Figure 2-6. These welds are used to join that member to the surface of another member that has been exposed through the hole.

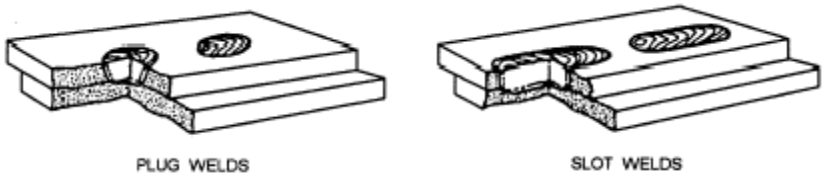


Figure 2-6: Plug and Slot Welds

2.4.5 Groove Welds

Groove welds (also may be referred to as Butt welds) are simply welds made in the groove between two members to be joined. The weld is adaptable to a variety of butt joints, as seen in Figure 2-7.

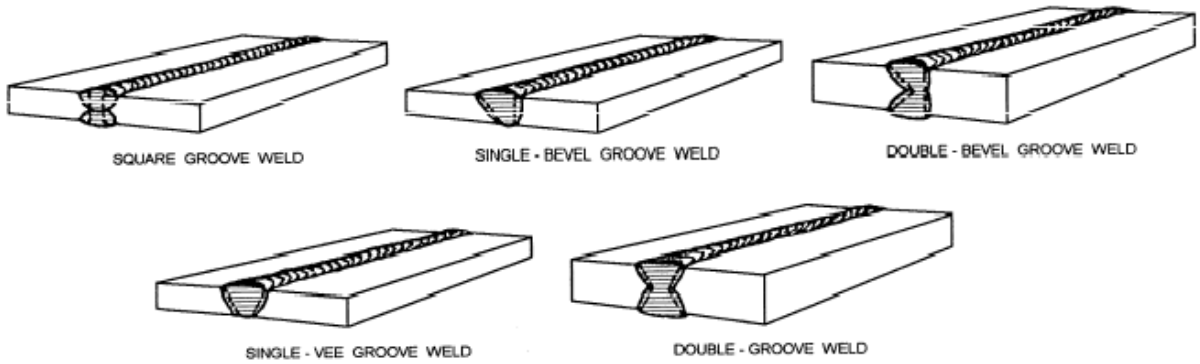


Figure 2-7: Types of Groove Welds

Groove welds may be joined with one or more weld beads, depending on the thickness of the metal. If two or more beads are deposited in the groove, the weld is made with multiple-pass layers. As a rule, a multiple-pass layer is made with stringer beads in manual operations.

The build-up sequence refers to the order in which the beads of a multiple-pass weld are deposited in the joint. Usually, before adding the next pass, the previous pass needs to cool down to a certain temperature which is called the inter-pass temperature. Also, before adding the next pass, the surface of the previous pass needs to be cleaned from slag, especially with SMAW, using a wire brush or other appropriate method.

2.5 Types of Welded Joints

The weld joint is where two or more metal parts are joined by welding. The five basic types of weld joints are the butt, corner, tee, lap, and edge.

2.5.1 Butt Joint

It is used to join two members aligned in the same plane – Figure 2-8 (a). This joint is frequently used in plate, sheet metal, and pipe work.

2.5.2 Corner and Tee Joints

These joints are used to join two members located at right angles to each other. In cross section, the corner joint forms an L-shape, and the tee joint has the shape of the letter T - Figure 2-8 (b) and (d).

2.5.3 Lap Joint

This joint is made by lapping one piece of metal over another - Figure 2-8 (e). This is one of the strongest types of joints available; however, for maximum joint efficiency, the overlap should be at least three times the thickness of the thinnest member of the joint.

2.5.4 Edge Joint

It is used to join the edges of two or more members lying in the same plane - Figure 2-8 (c). In most cases, one of the members is flanged, as seen in the figure. This type is frequently used in sheet metal work for joining metals 1/4 inch or less in thickness that are not subjected to heavy loads.

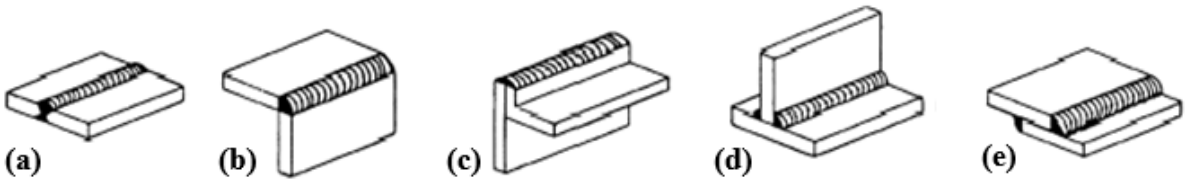


Figure 2-8: Types of Welded Joints

3.0 DRILLING

Drilling is the process of cutting holes in metals by using a drilling machine as shown in Figure 4-1. Drills are the tools used to cut away fine shavings of material as the drill advances in a rotational motion through the material.

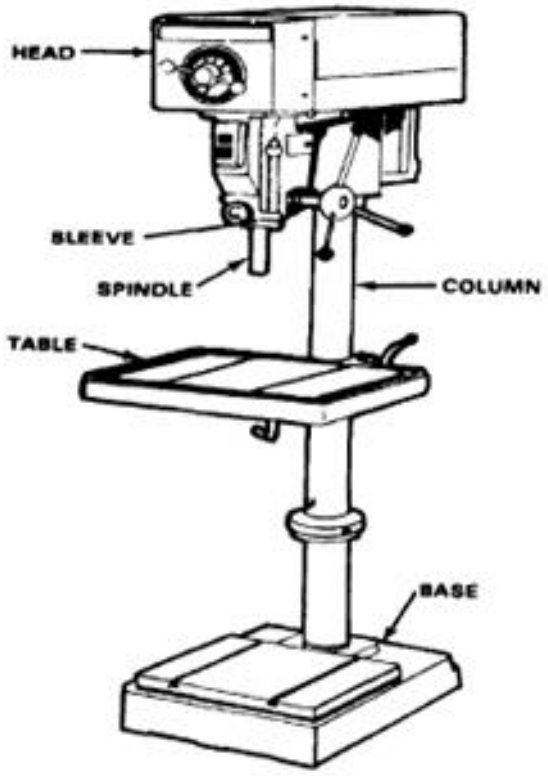


Figure 3-1: Drilling Machine

3.1 Twist Drill

The twist drill, as shown in Figure 4-2, is made of high speed steel. It is tempered to give maximum hardness throughout the parallel cutting portion. Flutes are incorporated to carry away the chips of metal and the outside surface is relieved to produce a cutting edge along the leading side of each flute.

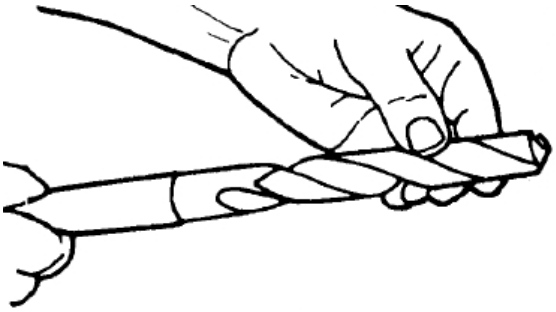


Figure 3-2: Twist Drill

3.2 Drill Features

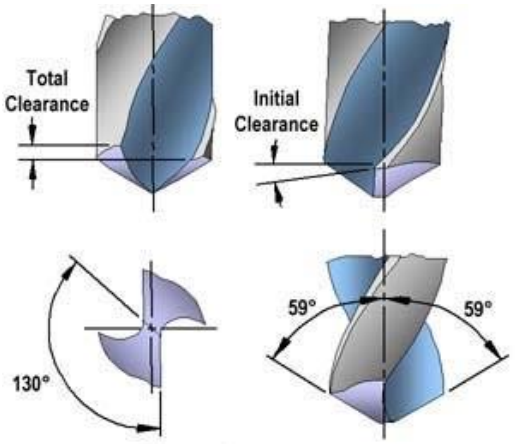


Figure 3-3: Drill Features

The point of the drill is ground to an angle of 59° to the centre line to give two equal cutting edges, and each side is ground back to give "relief" of about 12° to each cutting edge as shown in Figure 4-3.

3.3 Drill Operating Parameters

It is essential to select the correct cutting speed and feed. The following table shows the most common used cutting speed and feed rate.

Table 3-1: Cutting speed

Material	Cutting Speed
Mild steel	6 - 9 m/min
Stainless Steel	4 - 9 m/min
Aluminium	30 - 36 m/min

Table 3-2: Feed rate

Diameter of twist drill	Feed rate
5.5 mm	0.08 - 0.15 mm/rev
30 mm	0.04 - 0.55 mm/rev

3.4 Special Types of Drill

Countersink Drill, as shown in Figure 4-4, is used to form a conical shaped recess to enable a countersunk screw or bolt to fit flush with the surface of the work (Figure 4-5 b and c).

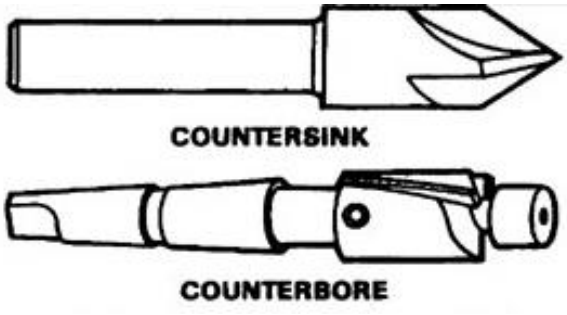


Figure 3-4: Countersink and Counterbore drills

Counterbore Drill, as shown in Figure 4-4, is used to form a flat, or cylindrical recess to accommodate the head of the bolt. It is also used to provide a level base on rough and curved surfaces for nuts and washers as in spotfacing (Figure 4-5a).

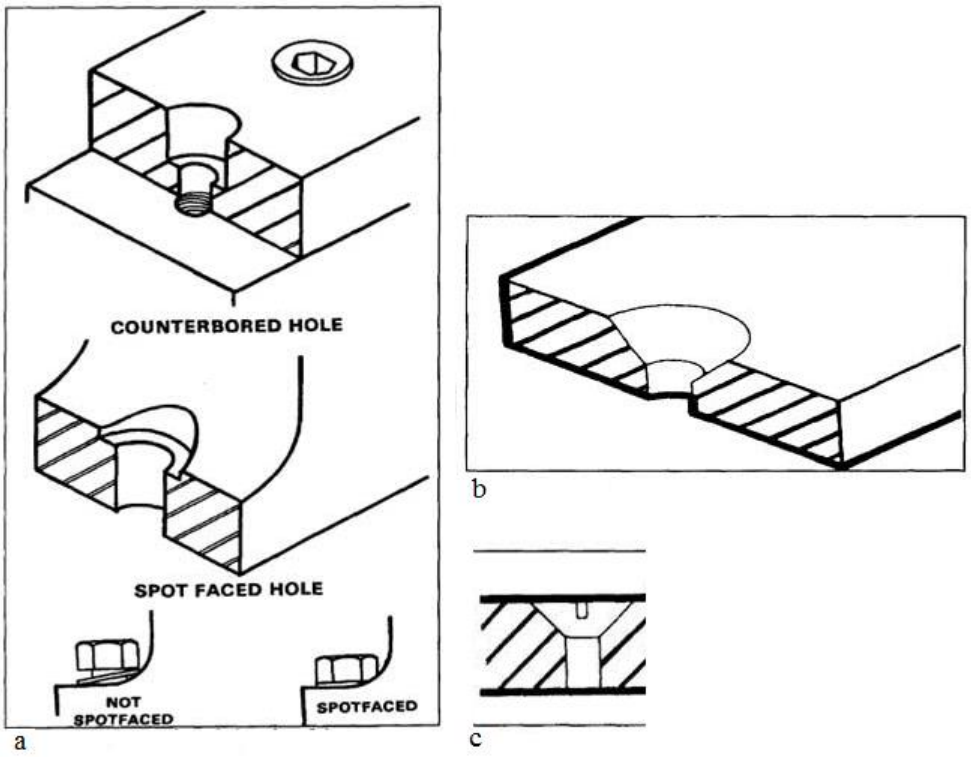


Figure 3-5: a. Counterbored and spotfaced holes, b. countersunk hole and c. countersunk hole with countersink screw

3.5 Safety and Care on Drilling

i. Twist drill must be clamped in the drill chuck tightly
ii. The workpiece to be drilled must be firmly secured by vice, or clamps.
iii. Drill guard, as shown in Figure 4-5, must be closed before switching on the machine.
iv. Use the correct drilling speed and apply suitable drilling force.
v. It is advisable to release the drill occasionally, lift the drill, and clear the hole of cutting.
vi. Apply cutting fluid in the cutting except for drilling Cast iron.
vii. Take care, when the drill has nearly penetrated through the workpiece.

4.0 REAMER AND THREADING TOOLS

4.1 Reamer

Functions of reamer are

1. to control the diameter of a hole
2. to improve the internal surface finish
3. to improve the roundness of the hole

Reamer is made of hardened High Carbon Steel or High Speed Steel. It is classified into hand reamer and machine reamer.

4.1.1 Hand Reamer

Solid hand reamers should be used when a greater accuracy in hole size is required. Hand reamer, as shown in Figure 4-6, has two types of flutes: - straight and spiral flutes. The spiral flutes hand reamer has a left hand spiral flutes. The purpose of the design is to prevent the reamer "screw in" the hole.

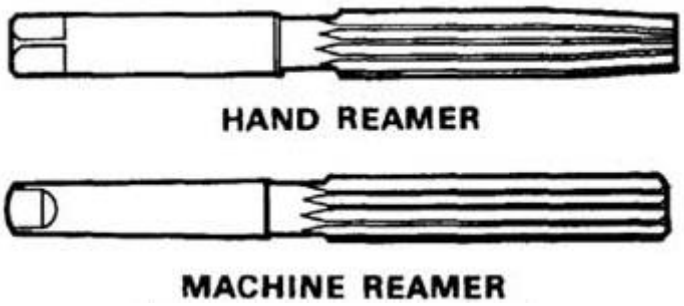


Figure 3-6: Hand and machine reamers

4.1.2 Machine Reamer

Machine reamer, as shown in Figure 4-6, has a straight shank or taper shank (Morse taper). The taper shank can fit directly into the spindle of a machine while the straight shank is hold by the collet.

4.1.3 Expanding Reamer/Adjustable Reamer

The cutting diameter can be slightly varied by adjusting an inner taper against the loss cutting blades as shown in Figure 4-7. This type is used primarily for repetitive work to maintain a consistent size throughout.



Figure 3-7: Adjustable Reamer

4.1.4 Safety, Precautions and Operation in Reaming

a. Care the sharp cutting edge especially in handling.

b. The amount of material to be removed by a reamer should be as small as possible, approximately 2-4% of diameter.

c. Reamer must only be turned in one direction, both cutting and removing the tools, otherwise the tool may jam.

d. Lubricant oil should be used except when cutting cast iron and brass.

e. Reaming can enlarge the size of hole, but cannot correct the position error in drilling.

4.2 Tap

Taps, as shown in Figure 4-8, are used to cut the internal screw threads. Taps are made of hardened High Carbon Steel or High Speed Steel. The ends of the shank are square to fit a wrench. Usually taps are provided in set of three - taper, second and plug tap.

4.2.1 Taper tap

This tap is tapered off for a length of 8 to 10 threads and is the first tap to be used in a hole to start the thread form. This feature gives the taper tap a very gradual cutting action that is less aggressive than that of the plug tap.

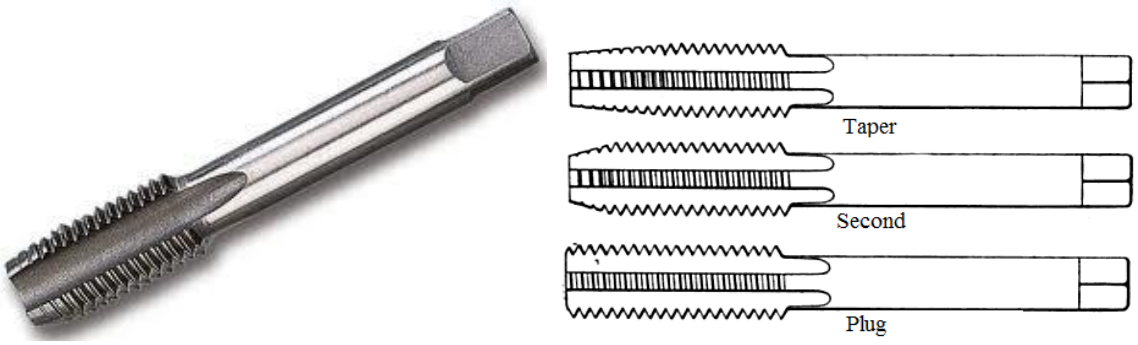


Figure 3-8: Tap and types of tap

4.2.2 Second tap

The tap is tapered off for a length of 3 to 5 threads to facilitate picking up the threads cut by the taper tap.

4.2.3 Plug or bottoming tap

This does not taper and is fully threaded throughout its length and is called a 'bottoming' tap. This tap is used to cut the bottom of a blind hole.



Figure 3-9: Tap Wrench

4.3 Precautions & operation in tapping

a. The size of the hole is important and the correct drill size should be determined from the handbook, standard table in the workshop or the recommendation on the shank of the tap.

b. Use taper tap first ensuring that it is kept square with top surface of work

c. Always use the correct size of wrench for the tap in use.

d. Lubricant oil should be used except when cutting cast iron and brass.

e. Use both hands to hold the wrench to maintain even torque.

f. About every half turn reverse action slightly to break the swarf and clear the threads.

g. When the tap reaches the bottom of the blind hole, care must be taken not to force as tap may break in the hole.

4.4 Die

Dies are used for cutting external threads on round bar or tubes. Dies are made of Hardened High Carbon Steel or High Speed Steel.

4.4.1 Split Die or Button Die

Split die is held in place in the stock as shown in Figure 4-10. The split permits a small amount of adjustment in the size of the die by adjusting the screws in the stock. Since split dies cut their thread complete in one cut, the die thread are tapered and backed off for one third of their length.

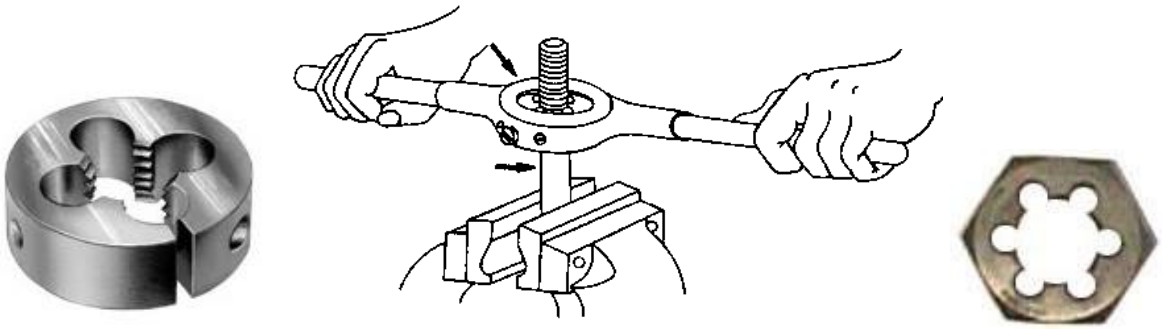


Figure 3-10: Slip die (left) and die nut (right)

4.4.2 Die nuts

Die nuts (Figure 4-10) are not capable of any adjustment. They are not usually employed for cutting threads from the bar, but for rectifying damage to existing threads. They are externally formed to hexagonal shape for use with a spanner.

4.4.2.1 Precautions and Operation of Die

a. The diameter of the blank rod must not larger than the outside diameter of thread to be cut.
b. Ensure that the die is set perpendicular to the rod.
c. Lubricant oil should be used except when cutting cast iron and brass.
d. About every half-turn reverse frequently to break the swarf otherwise the thread will tear.

5.0 MILLING

Milling is the process of machining flat, curved or irregular surfaces by feeding the workpiece against a rotating cutter containing a number of cutting edges. The milling machine consists basically of a motor driven spindle, which mounts and revolves the milling cutter and a reciprocating adjustable worktable, which mounts and feeds the workpiece.

5.1 Types of Milling Machines

Milling machines are basically classified as vertical or horizontal. These machines are also classified as knee-type, ram type, manufacturing or bed type and planer type.

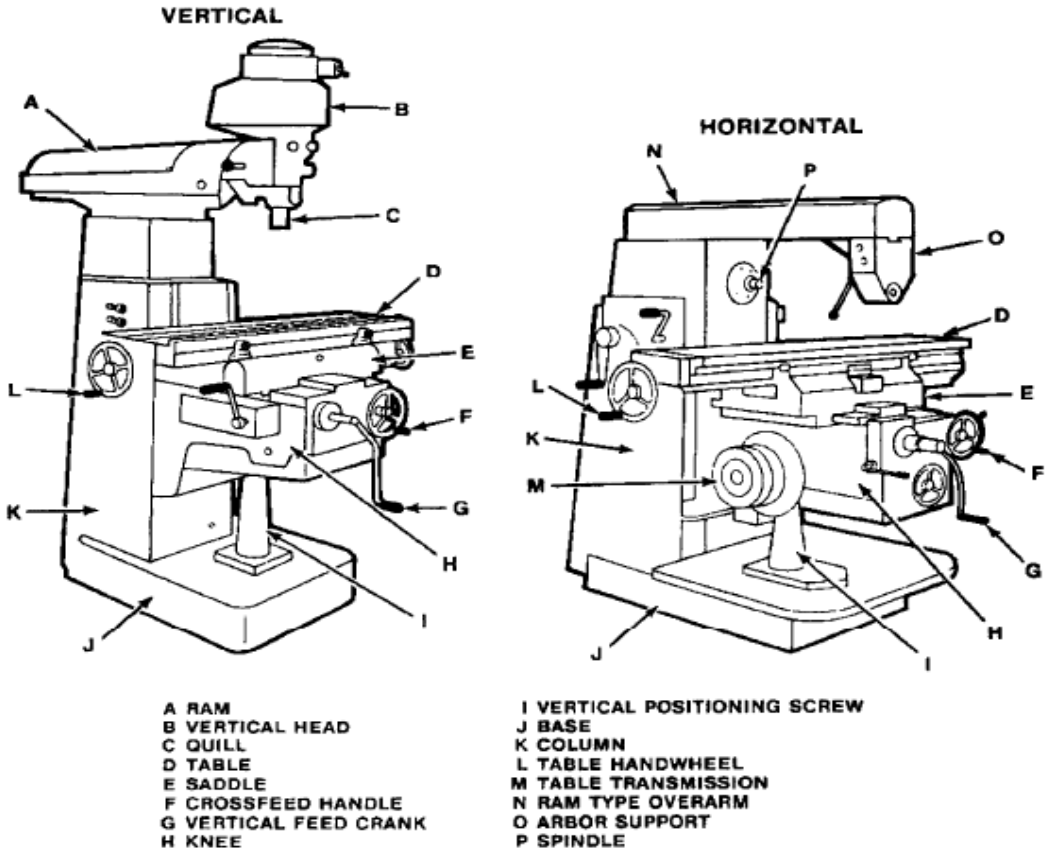


Figure 11: Basic Configuration of a Milling Machine

Most milling machines have self-contained electric drive motors, coolant system, variable spindle speeds and power operated table feeds. Figure 11 shows the basic milling machine configuration.

5.1.1 Knee-Type Milling Machine

Knee-type milling machines are characterized by a vertically adjustable worktable resting on a saddle which is supported by a knee. The knee is a massive casting that rides vertically on the milling machine column and can be clamped rigidly to the column in a position where the milling head and milling machine spindle are properly adjusted vertically for operation. The

plain vertical machines are characterized by a spindle located vertically, parallel to the column face, and mounted in a sliding head that can be fed up and down by hand or power.

Modern vertical milling machines are designed so the entire head can also swivel to permit working on angular surfaces. The turret and swivel head assembly is designed for making precision cuts and can be swung 360° on its base. Angular cuts to the horizontal plane may be made with precision by setting the head at any required angle within a 180° arc. The plain horizontal milling machine's column contains the drive motor and gearing and a fixed position horizontal milling machine spindle.

An adjustable overhead arm containing one or more arbor supports projects forward from the top of the column. The arm and arbor supports are used to stabilize long arbors. Supports can be moved along the overhead arm to support the arbor where support is desired depending on the position of the milling cutter or cutters. The milling machine's knee rides up or down the column on a rigid track. A heavy, vertical positioning screw beneath past the milling cutter.

The milling machine is excellent for forming flat surfaces, cutting dovetails and keyways, forming and fluting milling cutters and reamers, cutting gears, and so forth. Many special operations can be performed with the attachments available for milling machine use. The knee is used for raising and lowering. The saddle rests upon the knee and supports the worktable. The saddle moves in and out on a dovetail to control cross feed of the worktable. The worktable traverses to the right or left upon the saddle for feeding the workpiece past the milling cutter. The table may be manually controlled or power fed.

5.1.2 Universal Horizontal Milling Machine

The basic difference between a universal horizontal milling machine and a plain horizontal milling machine is the addition of a table swivel housing between the table and the saddle of the universal machine. This permits the table to swing up to 45° in either direction for angular and helical milling operations. The universal machine can be fitted with various attachments such as the indexing fixture, rotary table, slotting and rack cutting attachments, and various special fixtures.

5.1.3 Ram-Type Milling Machine

The ram-type milling machine is characterized by a spindle mounted to a movable housing on the column to permit positioning the milling cutter forward or rearward in a horizontal plane. Two popular ram-type milling machines are the universal milling machine and the swivel cutter head ram-type milling machine.

5.1.4 Universal Ram-Type Milling Machine

The universal ram-type milling machine is similar to the universal horizontal milling machine, the difference being, as its name implies, the spindle is mounted on a ram or movable housing.

5.1.5 Swivel Cutter Head Ram-Type Milling Machine

The cutter head containing the milling machine spindle is attached to the ram. The cutter head can be swivelled from a vertical spindle position to a horizontal spindle position or can be fixed at any desired angular position between vertical and horizontal. The saddle and knee are hand driven for vertical and cross feed adjustment while the worktable can be either hand or power driven at the operator's choice.

5.2 2.4 Safety Rules for Milling Machines

Milling machines require special safety precautions while being used.

a. Do not make contact with the revolving cutter.

b. Place a wooden pad or suitable cover over the table surface to protect it from possible damage.

c. Use the buddy system when moving heavy attachments.

d. Do not attempt to tighten arbor nuts using machine power.

e. When installing or removing milling cutters, always hold them with a rag to prevent cutting your hands.

f. While setting up work, install the cutter last to avoid being cut.

g. Never adjust the workpiece or work mounting devices when the machine is operating.

h. Chips should be removed from the workpiece with an appropriate rake and a brush.

i. Shut the machine off before making any adjustments or measurements.

j. When using cutting oil, prevent splashing by using appropriate splash guards. Cutting oil on the floor can cause a slippery condition that could result in operator injury

6.0 GRINDING

Grinding is the process of removing metal by the application of abrasives which are bonded to form a rotating wheel. When the moving abrasive particles contact the workpiece, they act as tiny cutting tools, each particle cutting a tiny chip from the workpiece. It is a common error to believe that grinding abrasive wheels remove material by a rubbing action; actually, the process is as much a cutting action as drilling, milling and lathe turning.

6.1 Type of Grinding Machines

The grinding machine supports and rotates the grinding abrasive wheel and often supports and positions the workpiece in proper relation to the wheel. The grinding machine is used for roughing and finishing flat, cylindrical, and conical surfaces; finishing internal cylinders or bores; forming and sharpening cutting tools; snagging or removing rough projections from castings and stampings; and cleaning, polishing, and buffing surfaces. Once strictly, a finishing machine or modem production grinding machines are used for complete roughing and finishing of certain classes of work.

From the simplest grinding machine to the most complex, grinding machines can be classified as utility grinding machines, cylindrical grinding machines and surface grinding machines. The average machinist will be concerned mostly with floor-mounted and bench-mounted utility grinding machines, buffing machines and reciprocating surface grinding machines.

6.1.1 Utility Grinding Machines

The utility grinding machine is intended for offhand grinding where the workpiece is supported in the hand and brought to bear against the rotating grinding abrasive wheel. The accuracy of this type of grinding machine depends on the operator's dexterity, skill and knowledge of the machine's capabilities and the nature of the work.

The utility grinding machine consists of a horizontally mounted motor with a grinding abrasive wheel attached to each end of the motor shaft. The electric-motor-driven machine is simple and common. It may be bench-mounted or floor-mounted. Generally, the condition and design of the shaft bearings as well as the motor rating determine the wheel size capacity of the machine. Suitable wheel guards and tool rests are provided for safety and ease of operation. Grinding machines come in various sizes and shapes as listed below.

6.1.1.1 Floor Mounted Utility Grinding Machine

The typical floor-mounted utility grinding machine stands waist-high and is secured to the floor by bolts. The floor-mounted utility grinding machine shown in Figure 12 mounts two 12-inch diameter by 2-inch-wide grinding abrasive wheels. The two wheel arrangement permits installing a coarse grain wheel for roughing purposes on one end of the shaft and a fine grain wheel for finishing purposes on the other end this saves the time that would be otherwise consumed in changing wheels. Each grinding abrasive wheel is covered by a wheel guard to increase the safety of the machine. Transparent eye shields, spark arresters and adjustable tool rests are provided for each grinding wheel. A tool tray and a water pan are mounted on the side of the base or pedestal. The water pan is used for quenching carbon steel cutting took as they are being ground. Using the 12-inch wheel, the machine provides a maximum cutting speed of approximately 5.500 SFPM. The 2-HP electric motor driving this machine has a maximum speed of 1.750 RPM.

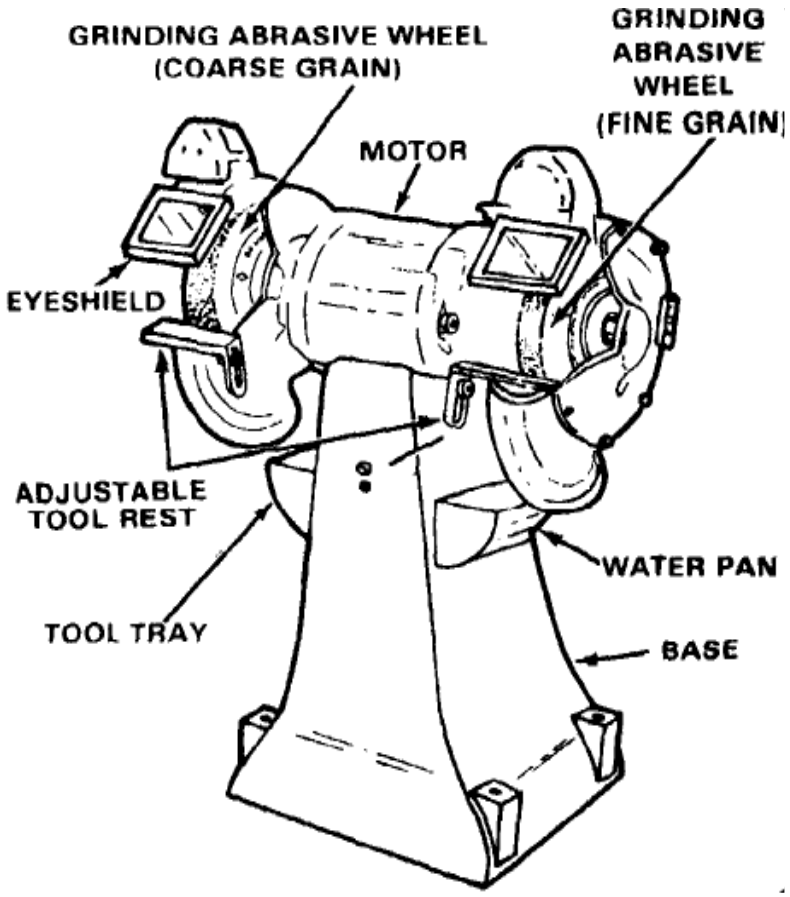


Figure 12: Floor-mounted Utility Grinding Machine

6.1.1.2 Bench Type Utility Grinding Machine

Like the floor mounted utility grinding machine, one coarse grinding wheel and one fine grinding wheel are usually mounted on the machine for convenience of operation. Each wheel is provided with an adjustable table tool rest and an eye shield for protection.

On this machine, the motor is equipped with a thermal over-load switch to stop the motor if excessive wheel pressure is applied thus preventing the burning out of the motor. The motor revolve at 3.450 RPM maximum to provide a maximum cutting speed for the 7 inch grinding wheels of about 6,300 surface feet per minute (SFPM). Figure 13 shows the diagram of a bench type utility grinding machine.

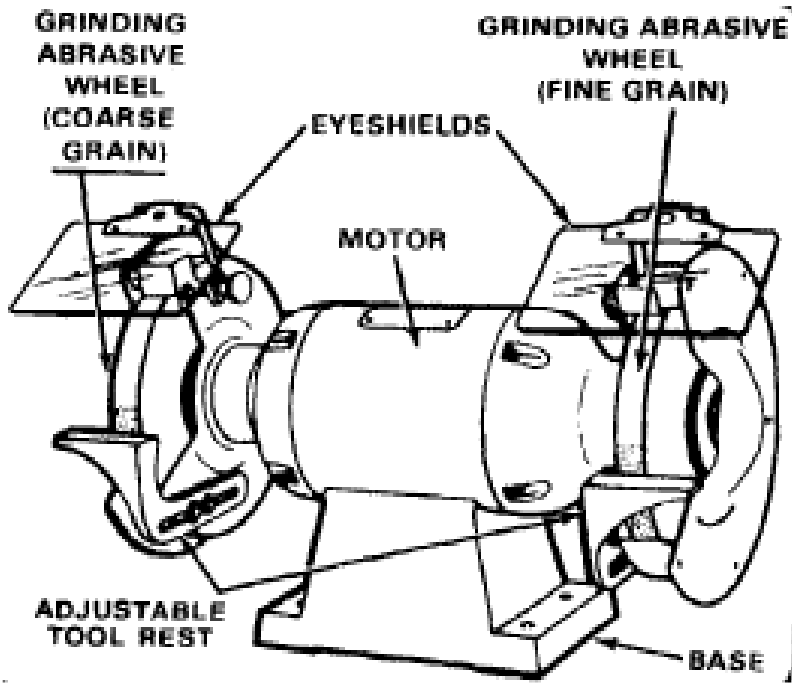


Figure 13: Bench type Utility Grinding Machine

6.1.1.3 Bench-Type Utility Grinding and Buffing Machine

The bench-type utility grinding and buffing machine is more suitable for miscellaneous grinding, cleaning, and buffing. It is not recommended for tool grinding since it contains no tool rests, eye shields, or wheel guards. This machine normally mounts a 4 inch-diameter wire wheel on one end. The wire wheel is used for cleaning and the abrasive wheel is used for general grinding. One of the two wheels can be removed and a buffing wheel mounted in its place for buffing and polishing. The 1/4-HP electric motor revolves at a maximum of 3,450 RPM. The maximum cutting speed of the 4-inch-diameter wheel is approximately 3,600 SFPM. Figure 14 shows the diagram of a bench type utility grinding and buffing machine.

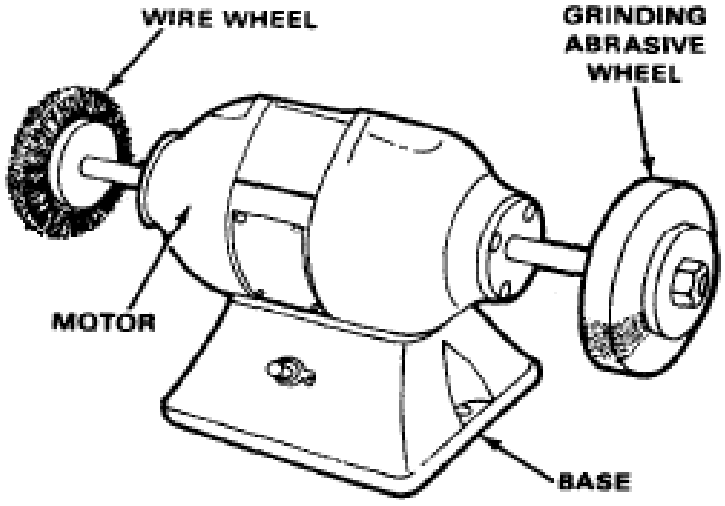


Figure 14: Bench type Utility Grinding and Buffing Machine

6.1.1.4 Bench-Type Utility Drill Grinding Machine

The bench-type drill grinding machine is intended for drill sharpening. The accuracy of this type of grinder is not dependent on the dexterity and skill of the operator because the drill is placed in a holding device. The holding device places the drill in the correct position for the clearance and included angle.

6.1.1.5 Bench-Type Tool and Cutter Grinder

The bench-type tool and cutter grinder (Figure 15) was designed primarily to grind end mills. It can also grind a large variety of small wood and steel cutters as well as slitting saw cutters up to 12 inches in diameter using the saw grinding attachment.

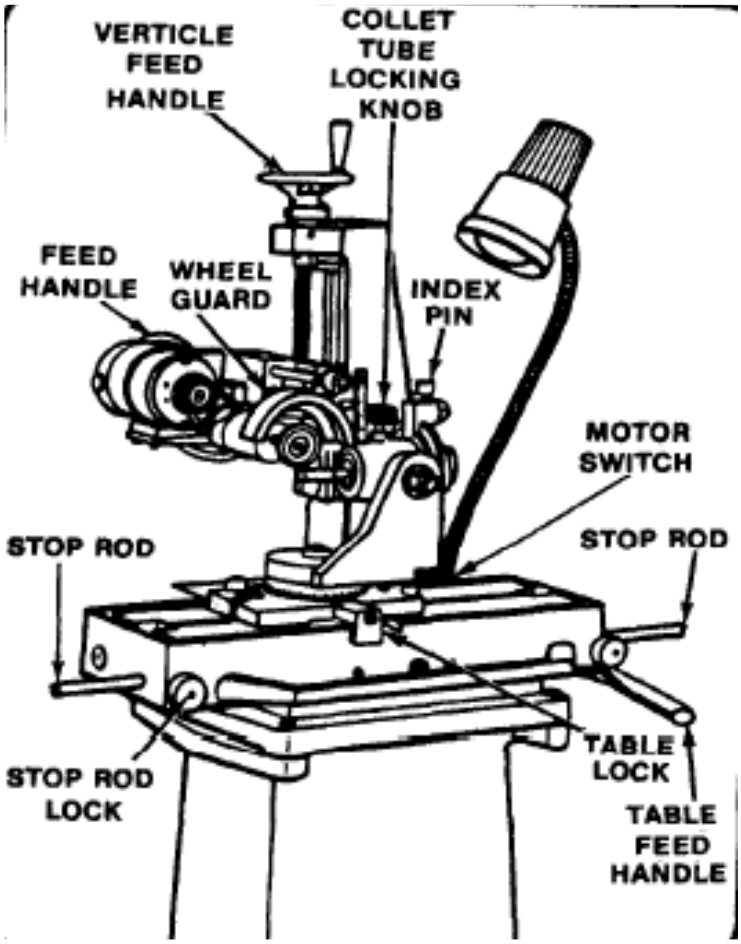


Figure 15: Bench- type Tool and Cutter Grinder

6.1.2 Reciprocating Surface Grinding Machine

The surface grinding machine is used for grinding flat surfaces. The workpiece is supported on a rectangular table which moves back and forth and reciprocates beneath the grinding wheel. Reciprocating surface grinding machines (Figure 16) generally have horizontal wheel spindles and mount straight or cylinder-type grinding abrasive wheels. Workpiece is fastened to the table and can be moved beneath the grinding abrasive wheel by hand or power feed. A magnetic chuck may be used for fastening the workpiece to the table. This grinding machine has an internal pump and piping network for automatic application and recirculation of a coolant to

the workpiece and wheel. The grinding abrasive wheel, mounted to the horizontal spindle is straight and cuts on its circumferential surface only. Grinding wheel speeds are adjustable.

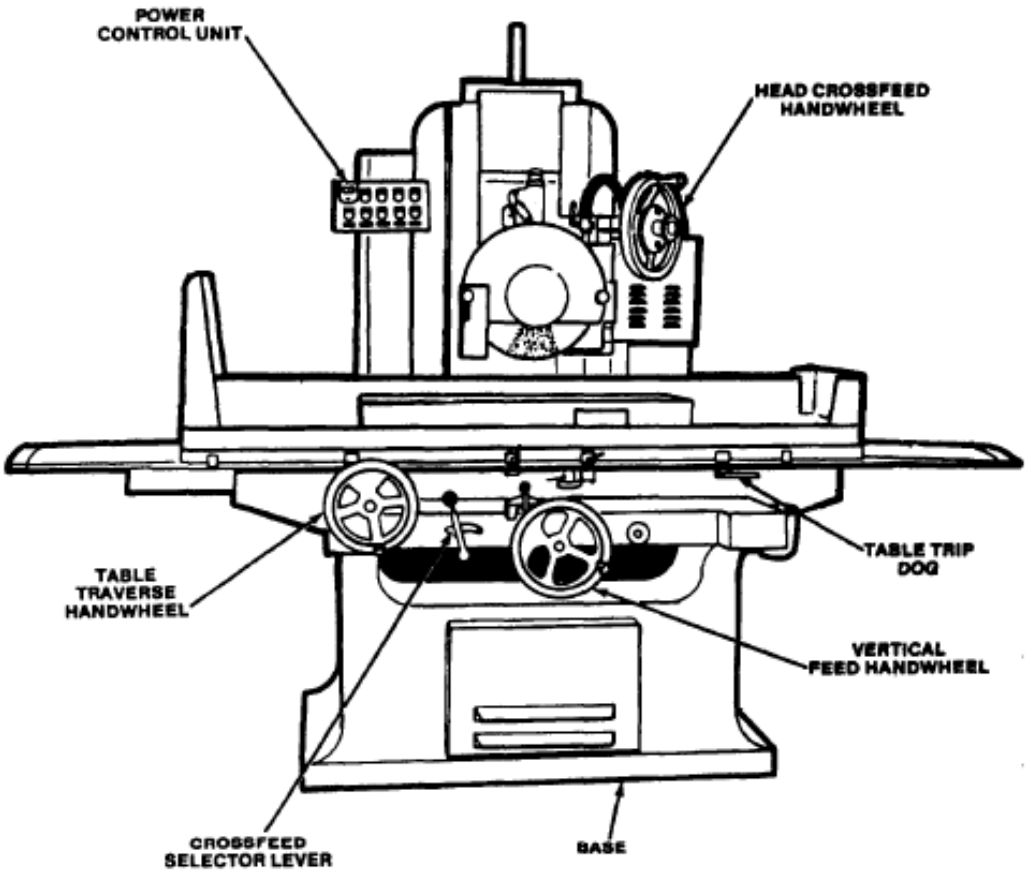


Figure 16: Bench- type Tool and Cutter Grinder

6.1.3 Tool Post Grinding Machine

The tool post grinding machine (Figure 16) is a machine tool attachment designed to mount to the tool post of engine lathes. It is used for internal and external grinding of cylindrical workplaces.

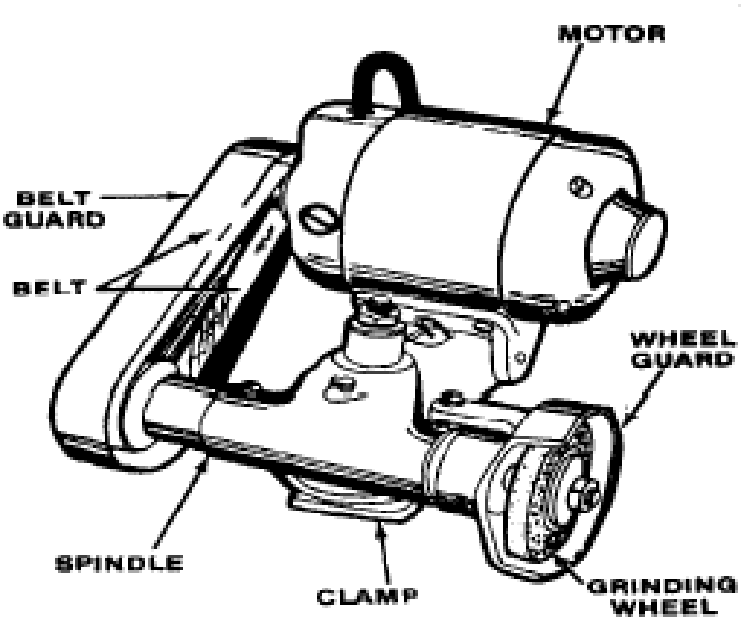


Figure 16: Tool Post Grinding Machine

6.2 Grinding Wheels

Grinding wheels come in many different sizes, shapes, and abrasives (Figure 17). Some of the various types are listed below.

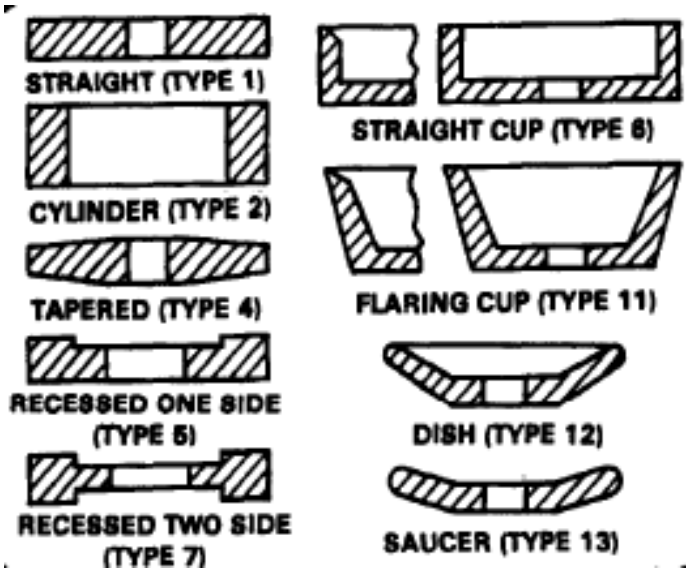


Figure 17: Standard Types of Grinding Wheels

6.2.1 Straight

Straight wheels, numbers 1, 5, and 7, are commonly applied to internal, cylindrical, horizontal spindle, surface, tool, and offhand grinding and snagging. The recesses in type numbers.5 and 7 accommodate mounting flanges. Type number 1 wheels from 0.006-inch to l/8-inch thick are used for cutting off stock and slotting.

6.2.2 Cylinder

Cylinder wheels, type number 2, may be arranged for grinding on either the periphery or side of the wheel.

6.2.3 Tapered

Tapered wheels, type number 4, take tapered safety flanges to keep pieces from flying if the wheel is broken while snagging.

6.2.4 Straight Cup

The straight cup wheel, type number 6, is used primarily for surface grinding, but can also be used for offhand grinding of flat surfaces. Plain or bevelled faces are available.

6.2.5 Flaring Cup

The flaring cup wheel, type number 11, is commonly used for tool grinding. With a resinoid bond, it is useful for snagging. Its face may be plain or bevelled.

6.2.6 Dish

The chief use of the dish wheel, type number 12, is in tool work. Its thin edge can be inserted into narrow places, and it is convenient for grinding the faces of form-relieved milling cutters and broaches.

6.2.7 Saucer

The saucer wheel, type number 13, is also known as a saw gummer because it is used for sharpening saws.

6.3 Abrasive Materials

The abrasive grains are the cutting tool of a grinding wheel. They actually cut small pieces or chips off the work as the wheel rotates. The shape of each grain is irregular with several sharp cutting edges. When these edges grow dull, the forces acting on the wheel tend to fracture the abrasive grains and produce new cutting edges.

6.3.1 Abrasives

Most grinding wheels are made of silicon carbide or aluminium oxide, both of which are artificial (manufactured) abrasives. Silicon carbide is extremely hard but brittle. Aluminium oxide is slightly softer but is tougher than silicon carbide. It dulls more quickly, but it does not

fracture easily therefore it is better suited for grinding materials of relatively high tensile strength.

6.3.2 Abrasive Grain Size

Abrasive grains are selected according to the mesh of a sieve through which they are sorted. For example, grain number 40 indicates that the abrasive grain passes through a sieve having approximately 40 meshes to the linear inch. A grinding wheel is designated coarse, medium, or fine according to the size of the individual abrasive grains making up the wheel.

6.3.3 Bonding Material

The abrasive particles in a grinding wheel are held in place by the bonding agent. The percentage of bond in the wheel determines, to a great extent, the "hardness" or "grade" of the wheel. The greater the percentage and strength of the bond, the harder the grinding wheel will be. "Hard" wheels retain the cutting grains longer, while "soft" wheels release the grains quickly. If a grinding wheel is "too hard" for the job, it will glaze because the bond prevents dulled abrasive particles from being released so new grains can be exposed for cutting. Besides controlling hardness and holding the abrasive, the bond also provides the proper safety factor at running speed. It holds the wheel together while centrifugal force is trying to tear it apart. The most common bonds used in grinding wheels are vitrified, silicate, shellac, resinoid, and rubber.

6.3.3.1 Vitrified

A vast majority of grinding wheels have a vitrified bond. Vitrified bonded wheels are unaffected by heat or cold and are made in a greater range of hardness than any other bond. They adapt to practically all types of grinding with one notable exception: if the wheel is not thick enough, it does not withstand side pressure as in the case of thin cut-off wheels.

6.3.3.2 Silicate

Silicate bond releases the abrasive grains more readily than vitrified bond. Silicate bonded wheels are well suited for grinding where heat must be kept to a minimum, such as grinding edged cutting tools. It is not suited for heavy-duty grinding. Thin cut-off wheels are sometimes made with a shellac bond because it provides fast cool cutting.

6.3.3.3 Resinoid

Resinoid bond is strong and flexible. It is widely used in snagging wheels (for grinding irregularities from rough castings), which operate at 9,500 SFPM. It is also used in cut-off wheels.

6.3.3.4 Rubber

In rubber-bonded wheels, pure rubber is mixed with sulphur. It is extremely flexible at operating speeds and permits the manufacture of grinding wheels as thin as 0.006 inch for slitting nibs. Most abrasive cut-off machine wheels have a rubber bond.

6.3.4 Grades of Hardness

The grade of a grinding wheel designates the hardness of the bonded material. Listed below are examples of those grades:

A soft wheel is one on which the cutting particles break away rapidly while a hard wheel is one on which the bond successfully opposes this breaking away of the abrasive grain.

Most wheels are graded according to hardness by a letter system. Most manufacturers of grinding abrasive wheels use a letter code ranging from A (very soft) to Z (very hard). Vitrified and silicate bonds usually range from very soft to very hard, shellac and resinoid bonds usually range from very soft to hard, and rubber bonds are limited to the medium to hard range.

The grade of hardness should be selected as carefully as the grain size. A grinding wheel that is too soft will wear away too rapidly, the abrasive grain will be discarded from the wheel before its useful life is realised. On the other hand, if the wheel is too hard for the job, the abrasive particles will become dull because the bond will not release the abrasive grain, and the wheel's efficiency will be impaired.

Figure 18 illustrates sections of three grinding abrasive wheels with different spacing of grains. If the grain and bond materials in each of these are alike in size and hardness, the wheel with the wider spacing will be softer than the wheel with the closer grain spacing. Thus, the actual hardness of the grinding wheel is equally dependent on grade of hardness and spacing of grains.

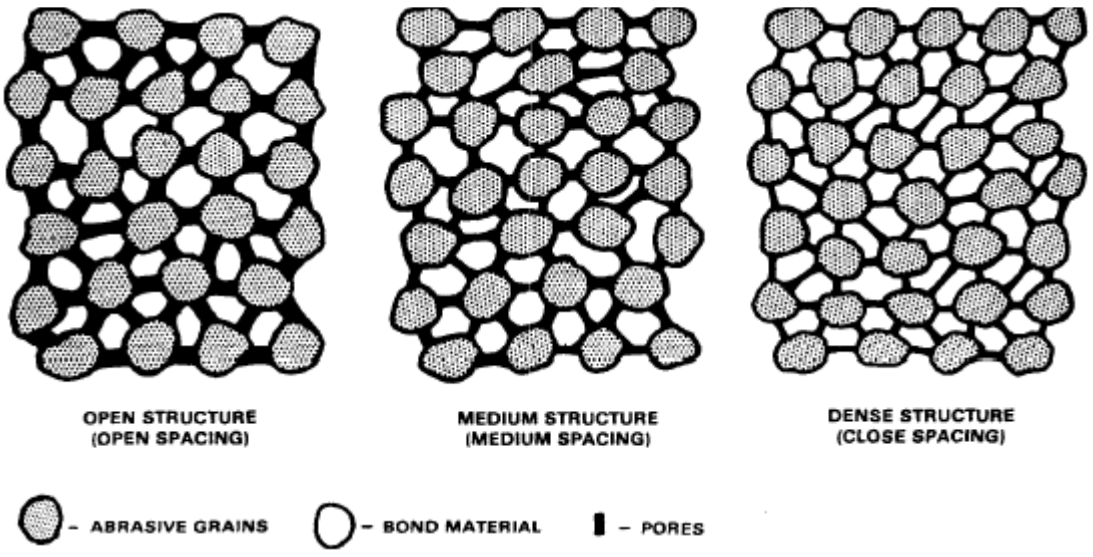


Figure 18: Grinding Wheel Abrasive

6.3.5 Abrasive Wheel Structure

Bond strength of a grinding wheel is not wholly dependent upon the grade of hardness but depends equally on the structure of the wheel, that is, the spacing of the grain or its density. The structure or spacing is measured in number of grains per cubic inch of wheel volume.

6.4 Markings on Grinding Wheels

Every grinding wheel is marked by the manufacturer with stencil or a small tag. The manufacturers have worked out a standard system of markings, shown in Figure 19.

For example use a wheel marked A36-L5-V23. The A refers to the abrasive which is Aluminium oxide. The 36 represents the grain size. The L shows the grade or degree of hardness, which is medium. The 5 refers to the structure of the wheel and the V refers to the bond type.

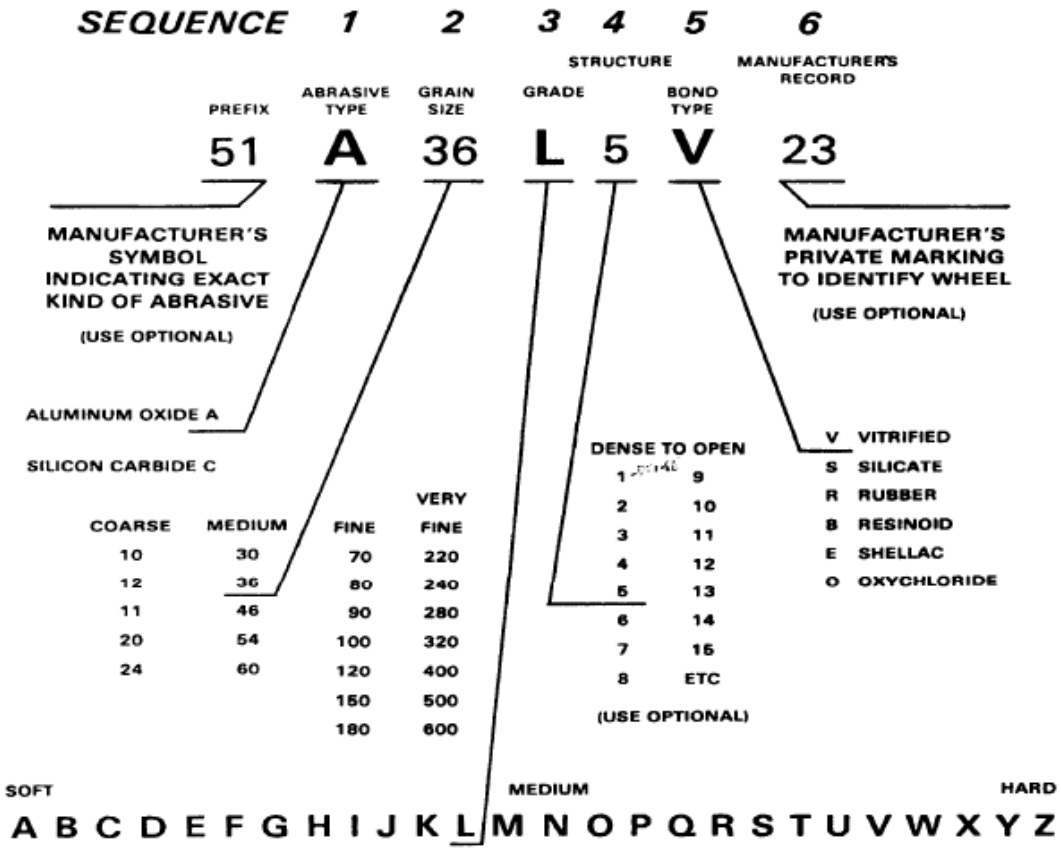


Figure 19: Standard System of Markings

6.5 **Grinding Speeds and Feeds**

In grinding, the speed of the grinding wheel in SFPM and the feed of the grinding wheel are as important as, and sometimes more important than, proper wheel selection. Occasionally, the grinder spindle should be checked with a tachometer to make sure it is running at its specified RPM. Too slow a speed will result in waste of abrasive, whereas an excessive speed will cause a hard grinding action and glaze the wheel, making the grinding inefficient. The feed of the

grinding wheel will determine to a certain extent the finish produced on the work and will vary for different types and shapes of grinding wheels.

Note:
If a wheel is permitted to exceed the maximum safe speed, it may disintegrate and cause injury to the operator and damage to the grinding machine.

4.3.1 Calculating Wheel Size or Speeds

Both cutting speeds in SFPM and rotational speed in RPM must be known to determine the size wheel to be used on a fixed-speed grinding machine. To determine the grinding wheel size, use the following formula:

$$D = \frac{12 \times SFPM}{RPM}$$

Where;
SFPM = Cutting speed of wheel (surface feet per minute)
RPM = Revolutions per minute of wheel
D = Calculated wheel diameter (inches)

7.0 SHAPING AND PLANING

Shaping and planing are processes for machining horizontal, vertical and inclined flat and contoured surfaces slots, grooves gears and other recesses means off surfaces, slots, grooves, gears, by special single-point tools.

Shaping and planing are similar operations. A straight, flat surface is created in both operations under low cutting speeds due to start-and-stop motion. Both use a single point cutting tool moved linearly relative to the workpiece.

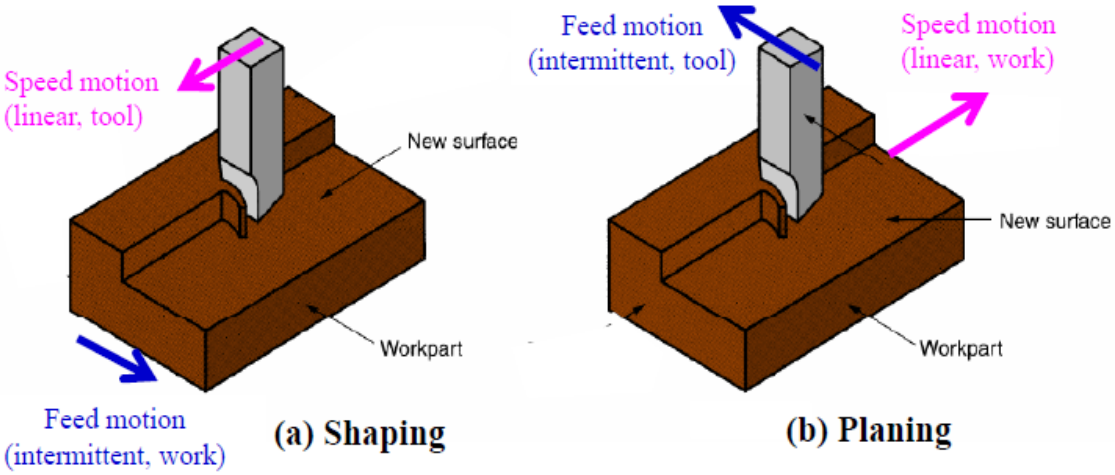


Figure 20: Differences between shaping and planing operations

In shaping, the work is cross-fed during the cut, and the speed motion is performed by the cutting tool - Figure 20 (a). On the other hand, in planing, the cutting tool is cross-fed, and the work part is moved past the tool in the speed motion - Figure 20 (b). Moreover, the shaping machine is smaller in size than the planing machine, which can machine heavy and long workpiece.

7.1 Shaping Machine

A photographic view of general configuration of shaping machine is shown in Figure 21(a). The main functions of shaping machines are to produce flat surfaces in different planes. The cutting motion provided by the linear forward motion of the reciprocating tool and the intermittent feed motion provided by the slow transverse motion of the job along with the bed result in producing a flat surface by gradual removal of excess material layer by layer in the form of chips. The vertical in-feed is given either by descending the tool holder or raising the bed or both. Straight grooves of various curved sections are also made in shaping machines by using specific form tools. The single point straight or form tool is clamped in the vertical slide which is mounted at the front face of the reciprocating ram whereas the workpiece is directly or indirectly through a vice is mounted on the bed. Figure 21 (b) shows the cutting tool of a shaping machine in action.

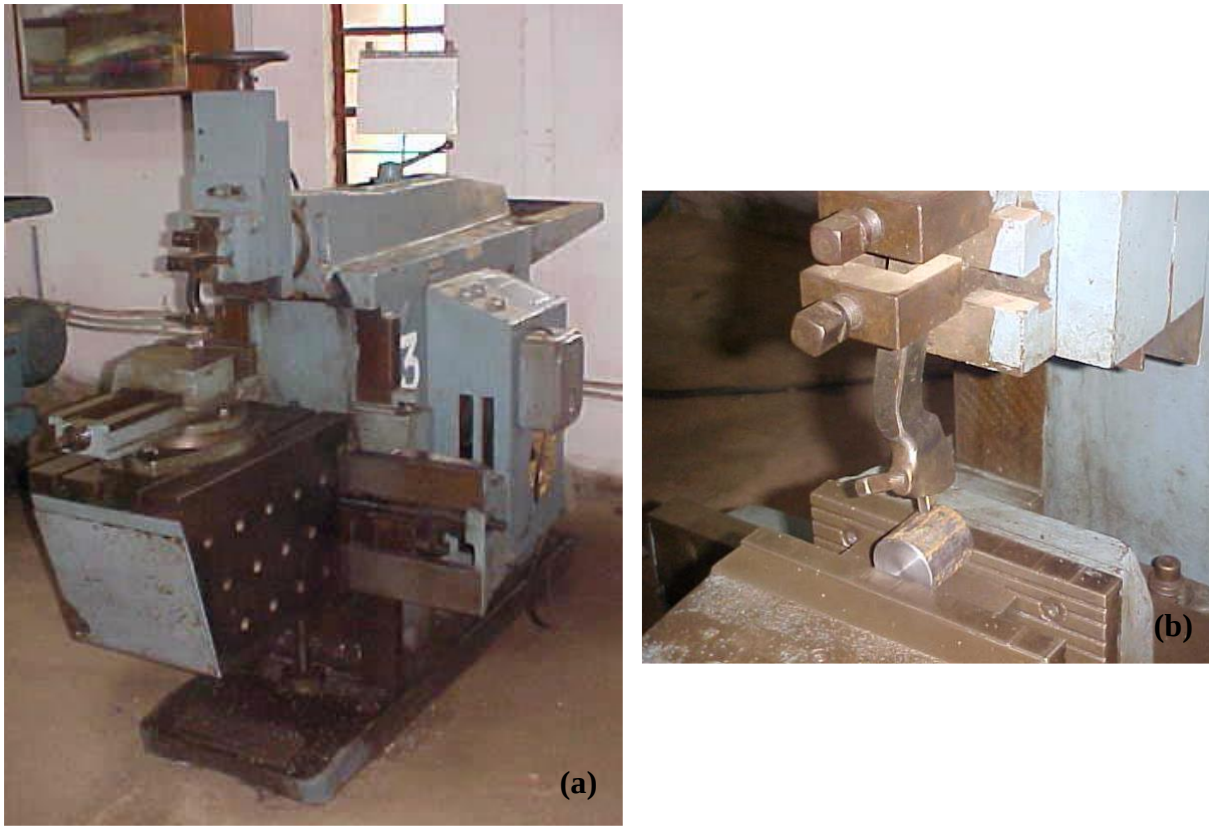


Figure 21: Photographic view of Shaping Machine

7.2 Planing Machine

Figure 22 (a) typically shows the general configuration of planing machine whilst Figure 21 (b) shows the cutting tool of planing machine in action. Like shaping machines, planing machines are also basically used for producing flat surfaces in different planes. However, the major differences between planing machines from shaping machines are:

- Though in principle both shaping and planing machines produce flat surface in the same way but in planing machine, instead of the tool, the workpiece reciprocates giving the fast cutting motion and instead of the job, the tool(s) is given the slow feed motion(s).

- Compared to shaping machines, planing machines are much larger and more rugged and generally used for large jobs with longer stroke length and heavy cuts. In planing machine, the workpiece is mounted on the reciprocating table and the tool is mounted on the horizontal rail which, again, can move vertically up and down along the vertical rails.

- Planing machines are more productive (than shaping machines) for longer and faster stroke, heavy cuts (high feed and depth of cut) possible and simultaneous use of a number of tools.

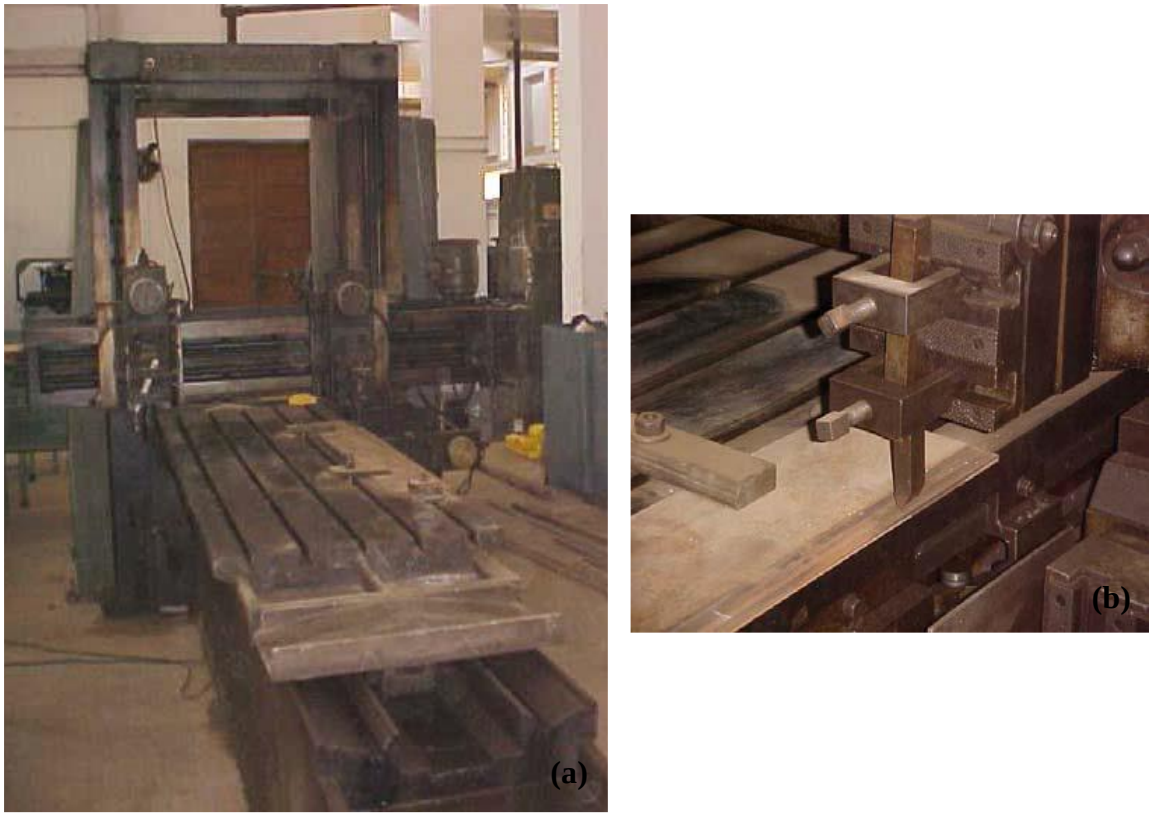


Figure 22: Photographic view of Planing Machine

As in shaping machines, in planing machines also;

- The length and position of stroke can be adjusted.
- Only single point tools are used.
- The quick return persists.
- Form tools are often used for machining grooves of curved section.
- Both shaping and planing machines can also produce large curved surfaces by using suitable attachments.

8.0 SHEET METAL FORMING

Sheet metal forming processes are those in which force is applied to a piece of sheet metal to modify its geometry rather than remove any material. The applied force stresses the metal beyond its yield strength, causing the material to plastically deform, but not to fail. By doing so, the sheet can be bent or stretched into a variety of complex shapes. Sheet metal forming processes include the following:

- Bending
- Roll forming
- Spinning
- Deep Drawing
- Stretch forming

8.1 Bending

Bending is a metal forming process in which a force is applied to a piece of sheet metal, causing it to bend at an angle and form the desired shape. A bending operation causes deformation along one axis, but a sequence of several different operations can be performed to create a complex part. Bent parts can be quite small, such as a bracket, or up to 20 feet in length, such as a large enclosure or chassis. A bend can be characterized by several different parameters, shown in Figure 33.

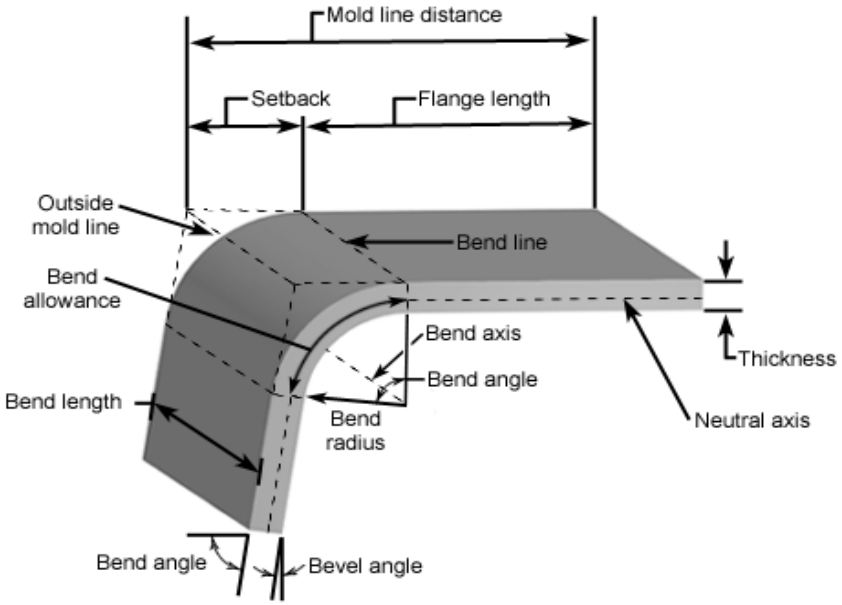


Figure 33: Bending Diagram

The act of bending results in both tension and compression in the sheet metal. The outside portion of the sheet will undergo tension and stretch to a greater length, while the inside portion experiences compression and shortens. The neutral axis is the boundary line inside the sheet metal, along which no tension or compression forces are present. As a result, the length of this axis remains constant.

8.1.1 Springback Factor

When bending a piece of sheet metal, the residual stresses in the material will cause the sheet to spring back slightly after the bending operation. Due to this elastic recovery, it is necessary

to over-bend the sheet a precise amount to achieve the desired bend radius and bend angle. The final bend radius will be greater than initially formed and the final bend angle will be smaller. The ratio of the final bend angle to the initial bend angle is defined as the springback factor, K_S. The amount of springback depends upon several factors, including the material, bending operation, and the initial bend angle and bend radius. Figure 34 shows the springback condition during bending.

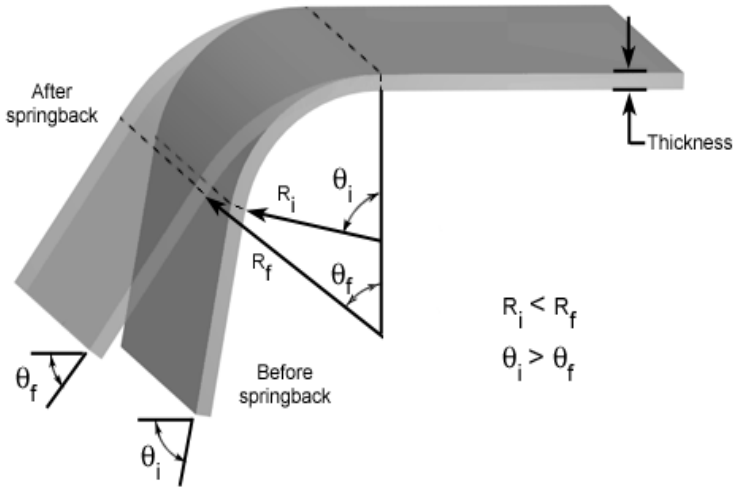


Figure 34: Springback condition

8.2 Bending Machine

Bending is typically performed on a machine called a press brake (Figure 35), which can be manually or automatically operated. For this reason, the bending process is sometimes referred to as press brake forming. Press brakes are available in a range of sizes (commonly 20-200 tons) in order to best suit the given application. A press brake contains an upper tool called the punch and a lower tool called the die, between which the sheet metal is located.

The sheet is carefully positioned over the die and held in place by the back gauge while the punch lowers and forces the sheet to bend. In an automatic machine, the punch is forced into the sheet under the power of a hydraulic ram. The bend angle achieved is determined by the depth to which the punch forces the sheet into the die. This depth is precisely controlled to achieve the desired bend. Standard tooling is often used for the punch and die, allowing a low initial cost and suitability for low volume production. Custom tooling can be used for specialized bending operations but will add to the cost.

The tooling material is chosen based upon the production quantity, sheet metal material and degree of bending. Naturally, a stronger tool is required to endure larger quantities, harder sheet metal and severe bending operations. In order of increasing strength, some common tooling materials include hardwood, low carbon steel, tool steel, and carbide steel.

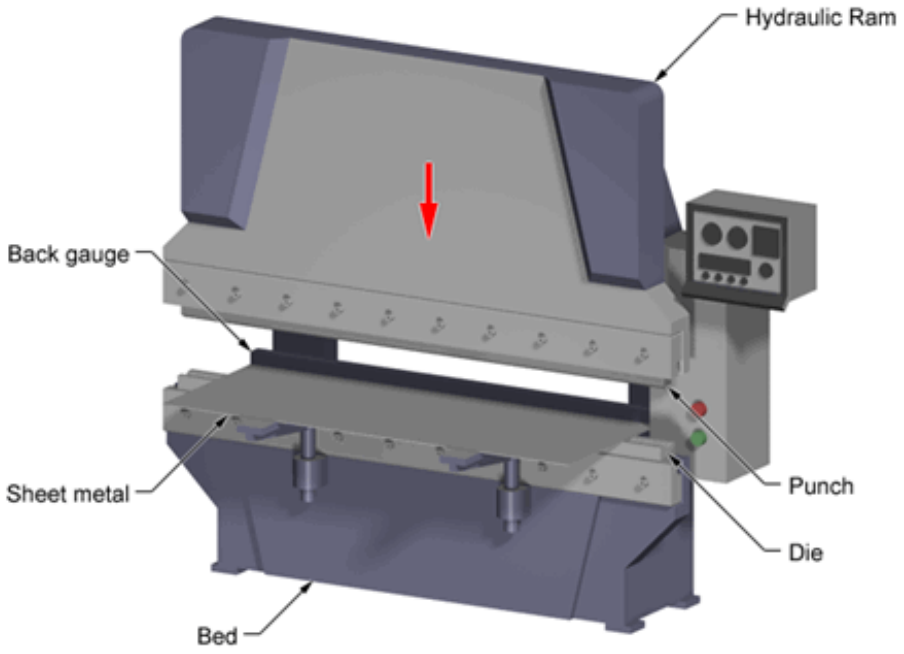
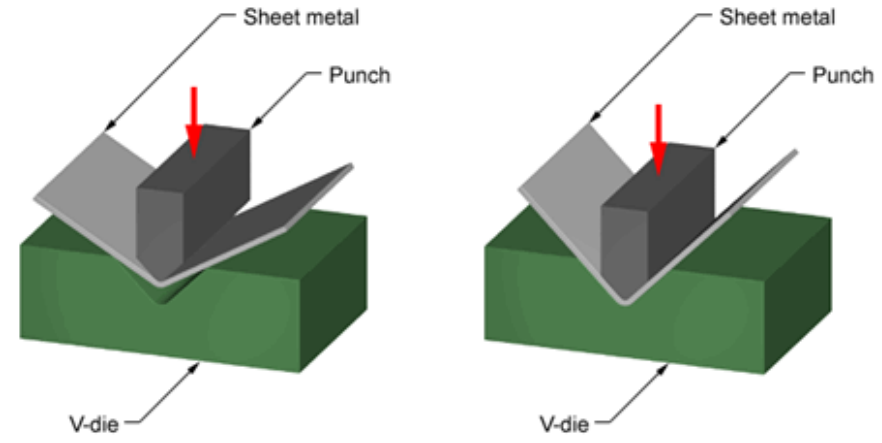


Figure 35: Press Brake

8.3 Bending Techniques

8.3.1 V-Bending

While using a press brake and standard die sets, there are still a variety of techniques that can be used to bend the sheet. The most common method is known as V-bending, in which the punch and die are "V" shaped. The punch pushes the sheet into the "V" shaped groove in the V-die, causing it to bend. If the punch does not force the sheet to the bottom of the die cavity, leaving space or air underneath, it is called "air bending" – Figure 36 (a). As a result, the V-groove must have a sharper angle than the angle being formed in the sheet. If the punch forces the sheet to the bottom of the die cavity, it is called "bottoming" – Figure 36 (b). This technique allows for more control over the angle because there is less springback. However, a higher tonnage press is required. In both techniques, the width of the "V" shaped groove, or die opening, is typically 6 to 18 times the sheet thickness. This value is referred to as the die ratio and is equal to the die opening divided by the sheet thickness.



(a) Air Bending **(b) Bottoming**

Figure 36: V-Bending Technique

8.3.2 Wipe Bending

In addition to V-bending, another common bending method is wipe bending (Figure 37), sometimes called edge bending. Wipe bending requires the sheet to be held against the wipe die by a pressure pad. The punch then presses against the edge of the sheet that extends beyond the die and pad. The sheet will bend against the radius of the edge of the wipe die.

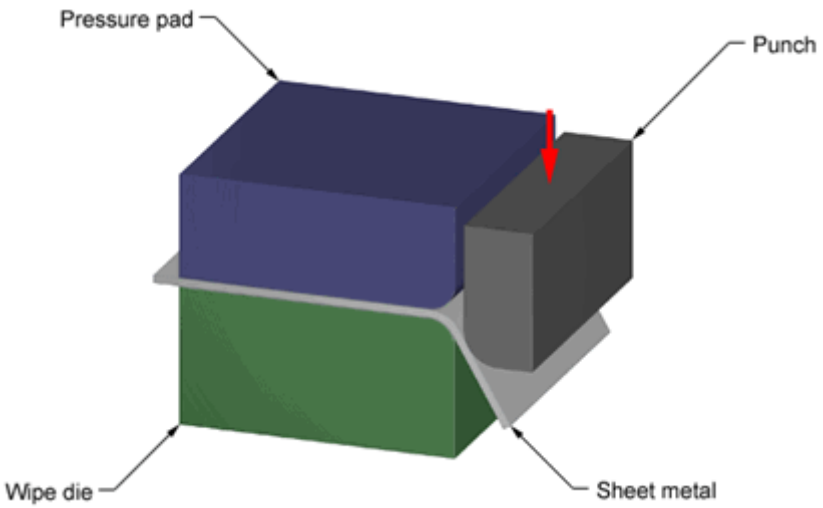


Figure 37: Wipe Bending Technique

1.0 Roll Forming

Roll forming is a metal forming process in which sheet metal is progressively shaped through a series of bending operations. The process is performed on a roll forming line in which the sheet metal stock is fed through a series of roll stations (Figure 38). Each station has a roller, referred to as a roller die, positioned on both sides of the sheet. The shape and size of the roller die may be unique to that station, or several identical roller dies may be used in different positions. The roller dies may be above and below the sheet, along the sides, at an angle, etc. As the sheet is forced through the roller dies in each roll station, it plastically deforms and bends. Each roll station performs one stage in the complete bending of the sheet to form the desired part. The roller dies are lubricated to reduce friction between the die and the sheet, thus reducing the tool wear. Also, lubricant can allow for a higher production rate, which will also depend on the material thickness, number of roll stations, and radius of each bend. The roll

forming line can also include other sheet metal fabrication operations before or after the roll forming, such as punching or shearing.

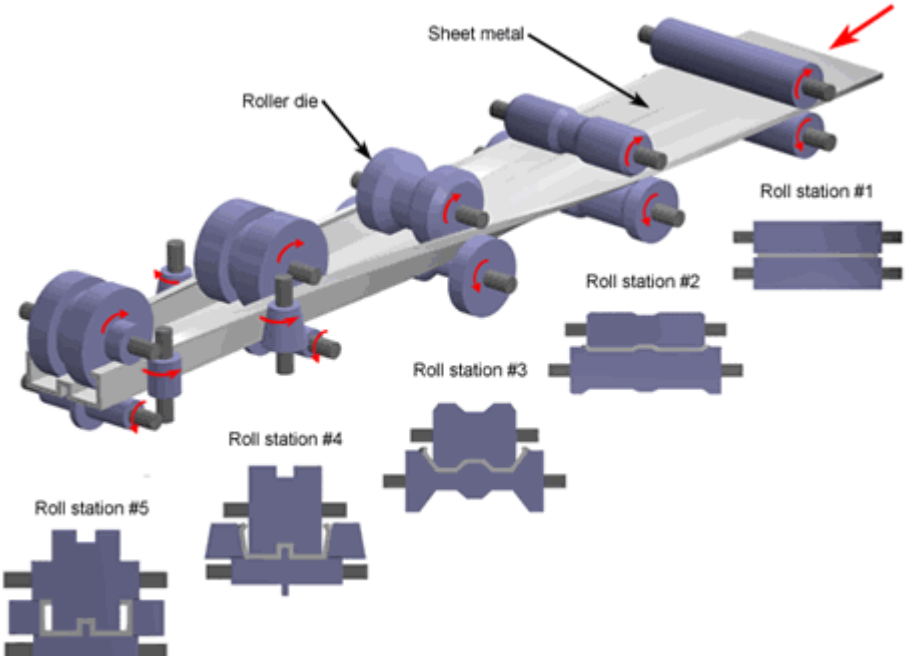


Figure 38: Roll Forming Line

The roll forming process can be used to form a sheet into a wide variety of cross-section profiles. An open profile is most common, but a closed tube-like shape can be created as well. Because the final form is achieved through a series of bends, the part does not require a uniform or symmetric cross-section along its length. Roll forming is used to create very long sheet metal parts with typical widths of 1-20 inches and thicknesses of 0.004-0.125 inches. However wider and thicker sheets can be formed, some up to 5 ft. wide and 0.25 inches thick. The roll forming process is capable of producing parts with tolerances as tight as ±0.005 inches. Typical roll formed parts include panels, tracks, shelving, etc. These parts are commonly used in industrial and commercial buildings for roofing, lighting, storage units, and HVAC applications.

2.0 Spinning

Spinning, sometimes called spin forming, is a metal forming process used to form cylindrical parts by rotating a piece of sheet metal while forces are applied to one side. A sheet metal disc is rotated at high speeds while rollers press the sheet against a tool, called a mandrel, to form the shape of the desired part. Spun metal parts have a rotationally symmetric, hollow shape, such as a cylinder, cone, or hemisphere. Examples include cookware, hubcaps, satellite dishes, rocket nose cones, and musical instruments.

Spinning is typically performed on a manual or CNC lathe and requires a blank, mandrel, and roller tool (Figure 39). The blank is the disc-shaped piece of sheet metal that is pre-cut from sheet stock and will be formed into the part. The mandrel is a solid form of the internal shape of the part, against which the blank will be pressed. For more complex parts, such as those with re-entrant surfaces, multi-piece mandrels can be used.

Because the mandrel does not experience much wear in this process, it can be made from wood or plastic. However, high volume production typically utilizes a metal mandrel. The mandrel and blank are clamped together and secured between the headstock and tailstock of the lathe to

be rotated at high speeds by the spindle. While the blank and mandrel rotate, force is applied to the sheet by a tool, causing the sheet to bend and form around the mandrel. The tool may make several passes to complete the shaping of the sheet. This tool is usually a roller wheel attached to a lever. Rollers are available in different diameters and thicknesses and are usually made from steel or brass. The rollers are inexpensive and experience little wear allowing for low volume production of parts.

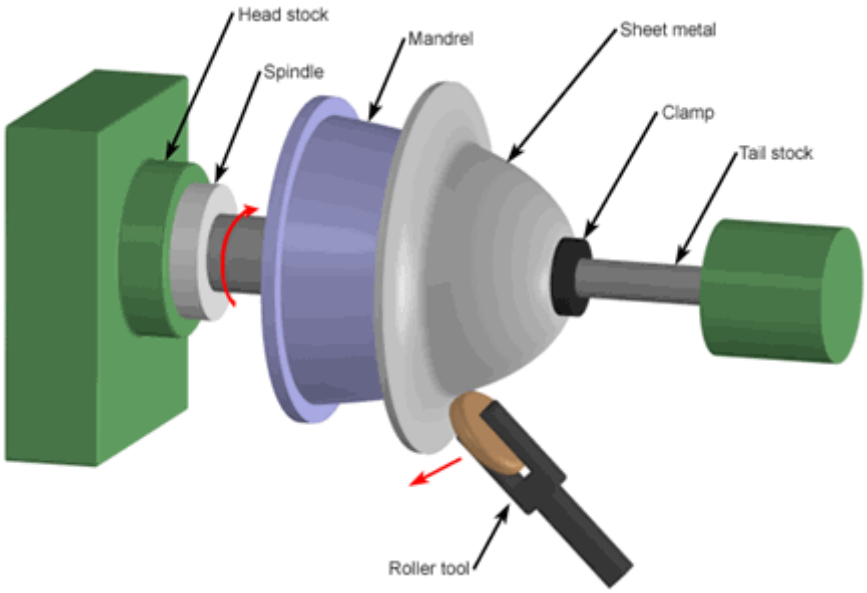
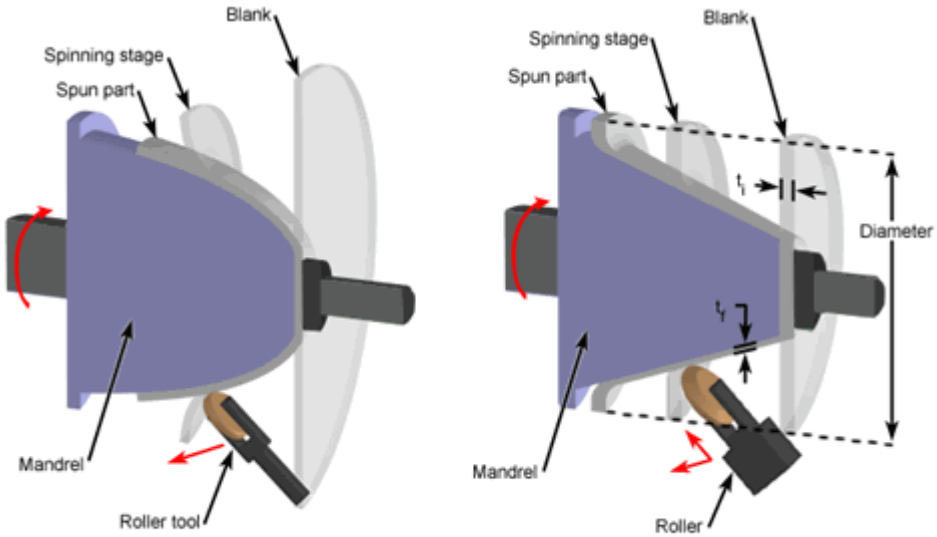


Figure 39: Spinning Lathe

2.1 Spinning Techniques

There are two distinct spinning methods, referred to as conventional spinning and shear spinning. In conventional spinning, the roller tool pushes against the blank until it conforms to the contour of the mandrel - Figure 40 (a). The resulting spun part will have a diameter smaller than the blank, but will maintain a constant thickness. In shear spinning, the roller not only bends the blank against the mandrel, it also applies a downward force while it moves, stretching the material over the mandrel - Figure 40 (b). By doing so, the outer diameter of the spun part will remain equal to the original blank diameter, but the thickness of the part walls will be thinner.



(a) Conventional Spinning **(b)** Shear Spinning

Figure 40: Spinning Techniques

3.0 Deep Drawing

Deep drawing is a metal forming process in which sheet metal is stretched into the desired part shape. A tool pushes downward on the sheet metal, forcing it into a die cavity in the shape of the desired part. The tensile forces applied to the sheet cause it to plastically deform into a cup-shaped part. Deep drawn parts are characterized by a depth equal to more than half of the diameter of the part. These parts can have a variety of cross sections with straight, tapered, or even curved walls, but cylindrical or rectangular parts are most common. Deep drawing is most effective with ductile metals, such as aluminium, brass, copper, and mild steel. Examples of parts formed with deep drawing include automotive bodies and fuel tanks, cans, cups, kitchen sinks, and pots and pans. Figure 41 depicts a deep drawing process.

The deep drawing process requires a blank, blank holder, punch, and a die. The blank is a piece of sheet metal, typically a disc or rectangle, which is pre-cut from stock material and will be formed into the part. The blank is clamped down by the blank holder over the die, which has a cavity in the external shape of the part. A tool called a punch moves downward into the blank and draws, or stretches, the material into the die cavity. The movement of the punch is usually hydraulically powered to apply enough force to the blank. Both the die and punch experience wear from the forces applied to the sheet metal and are therefore made from tool steel or carbon steel.

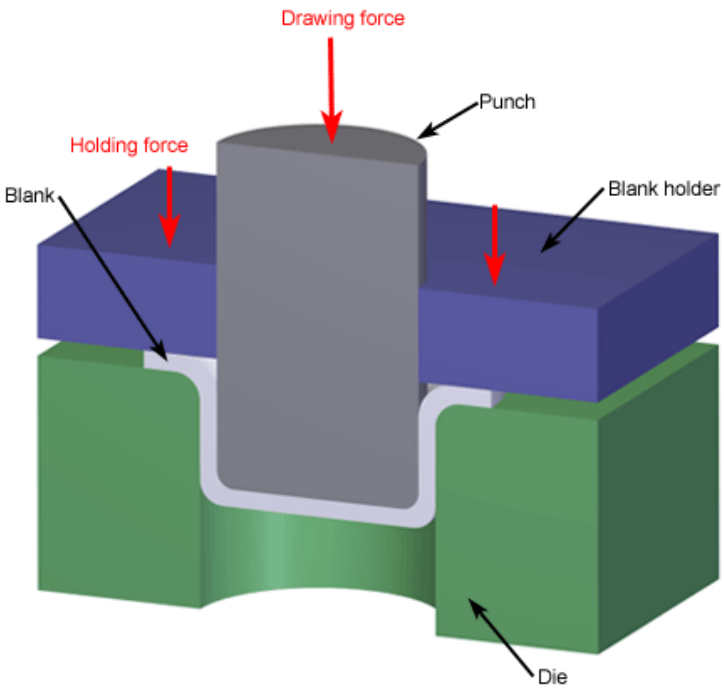


Figure 41: Deep Drawing

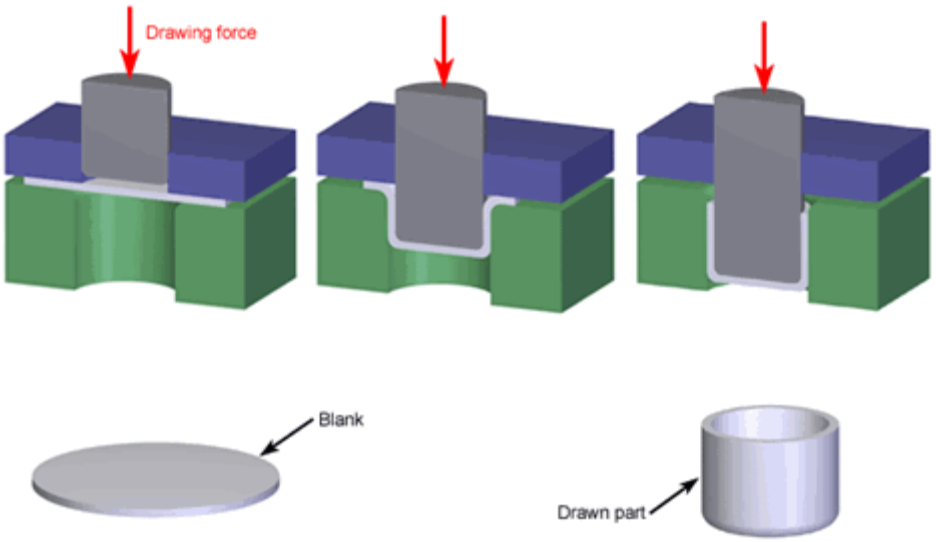


Figure 42: Deep Drawing Sequence

The process of drawing the part sometimes occurs in a series of operations, called draw reductions (Figure 42). In each step, a punch forces the part into a different die, stretching the part to a greater depth each time. After a part is completely drawn, the punch and blank holder can be raised and the part removed from the die. The portion of the sheet metal that was clamped under the blank holder may form a flange around the part that can be trimmed off.

4.0 Stretch Forming

Stretch forming is a metal forming process in which a piece of sheet metal is stretched and bent simultaneously over a die in order to form large contoured parts (Figure 43). Stretch forming is performed on a stretch press, in which a piece of sheet metal is securely gripped along its edges by gripping jaws. The gripping jaws are each attached to a carriage that is pulled by pneumatic or hydraulic force to stretch the sheet. The tooling used in this process is a stretch form block, called a form die, which is a solid contoured piece against which the sheet metal will be pressed.

The most common stretch presses are oriented vertically, in which the form die rests on a press table that can be raised into the sheet by a hydraulic ram. As the form die is driven into the sheet, which is gripped tightly at its edges, the tensile forces increase and the sheet plastically deforms into a new shape. Horizontal stretch presses mount the form die sideways on a stationary press table, while the gripping jaws pull the sheet horizontally around the form die.

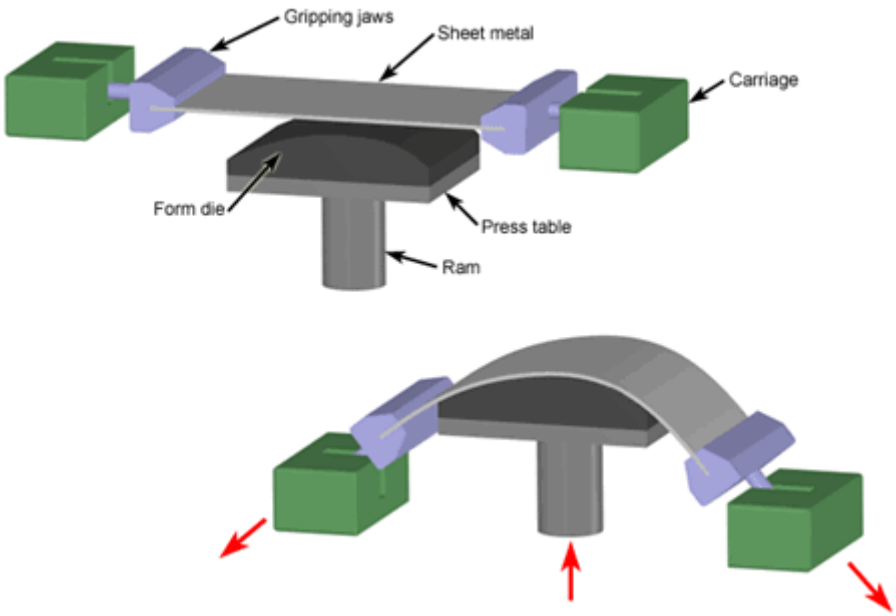


Figure 43: Stretch Forming

Stretch formed parts are typically large and possess large radius bends. The shapes that can be produced vary from a simple curved surface to complex non-uniform cross sections. Stretch forming is capable of shaping parts with very high accuracy and smooth surfaces. Ductile materials are preferable, the most commonly used being aluminium, steel, and titanium. Typical stretch formed parts are large curved panels such as door panels in cars or wing panels on aircraft. Other stretch formed parts can be found in window frames and enclosures.